



D

STM Exercise 5

Phase Diagrams

(full solubility in the solid state)

(no solubility in the solid state)

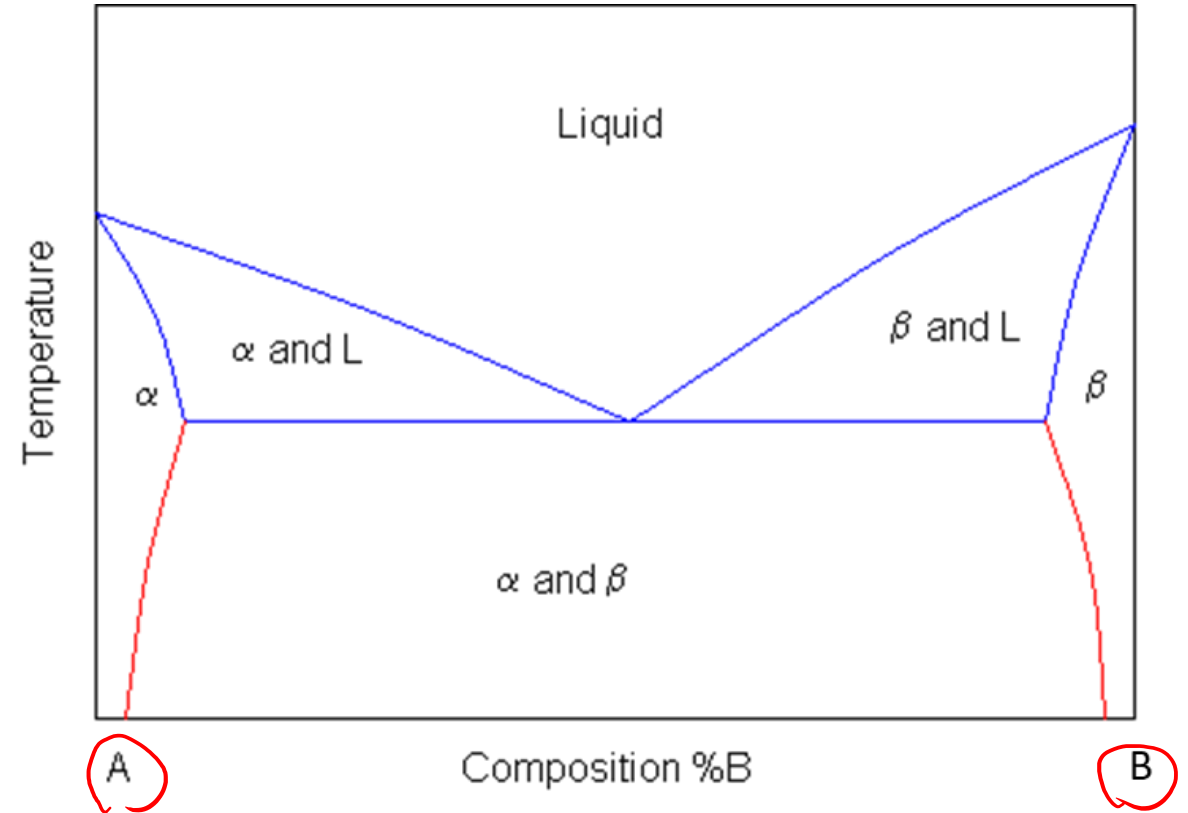
Phase Diagram

it shows the equilibrium states of a mixture, so that given a temperature and composition, it is possible to calculate which phases will be formed, and in what quantities.

Where **L** stands for liquid, and **A** and **B** are the two components and **α** and **β** are two solid phases rich in A and B respectively.

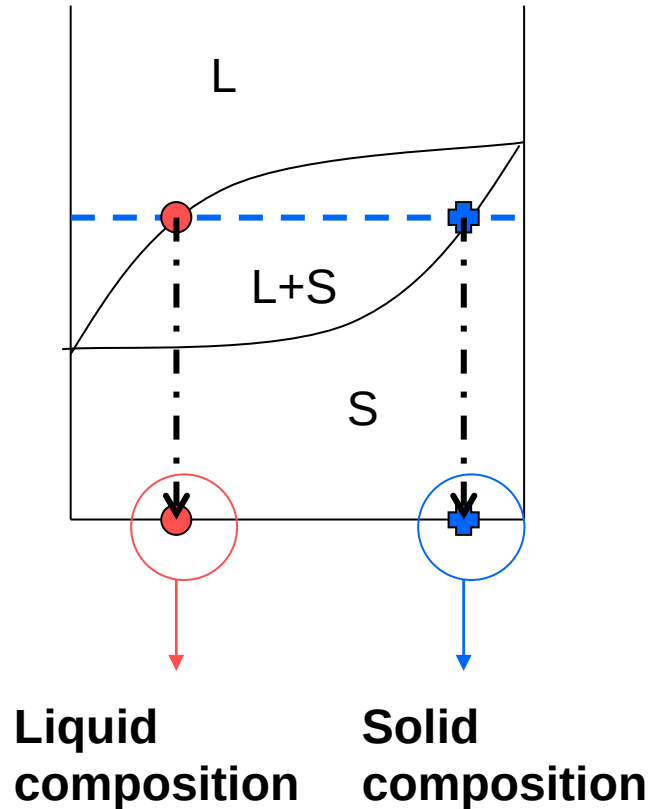
The blue lines represent the liquidus and solidus lines

The red lines involve a solid-to-solid transition (solvus lines)



Determination of the compositions of phases in equilibrium in a biphasic zone:

HORIZONTAL RULE



➤ Draw the tie line - - - - -

➤ Individuate intersections with **liquidus** (●) and **solidus** (⊕) lines

➤ The **composition of the liquid** is the value on the x axis correspondent to the intersection between the tie line and the liquidus line.

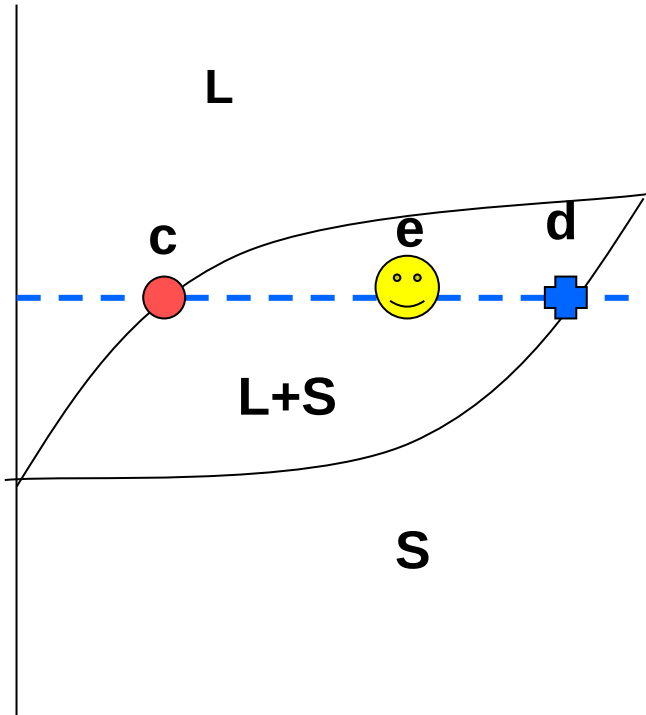
➤ The **composition of the solid** is the value on the x axis correspondent to the intersection of the tie line with the solidus line.

LIQUIDUS CURVE = above it only liquid is present

SOLIDUS CURVE = below it only solid is present

Phases relative amounts determination in a biphasic zone, at equilibrium:

LEVER RULE



- Draw the tie line - - - - -
- Individuate the intersection between tie line and liquidus line (• c) and solidus line (⊕ d)
- The % of phases in a point of the biphasic region (e) is defined as:

$$\%L = (ed/cd) \cdot 100$$

$$\%S = (ce/cd) \cdot 100$$

The lever rule

If an alloy consists of more than one phase, the amount of each phase present can be found by applying **the lever rule** to the phase diagram.

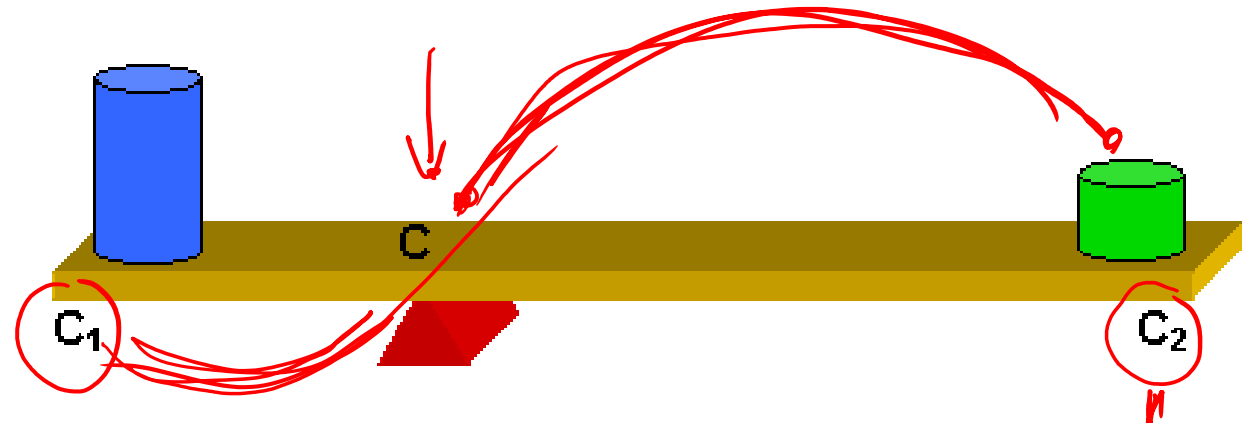
The lever rule can be explained by considering a simple balance.

The composition of the alloy is represented by the fulcrum C, and the compositions of the two phases by the ends of a bar (C₁ and C₂).

Each phase are determined by the weights needed to balance the system.

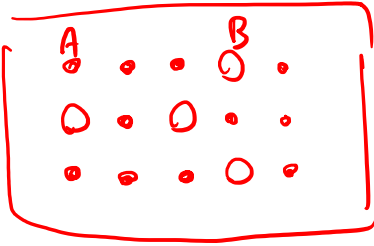
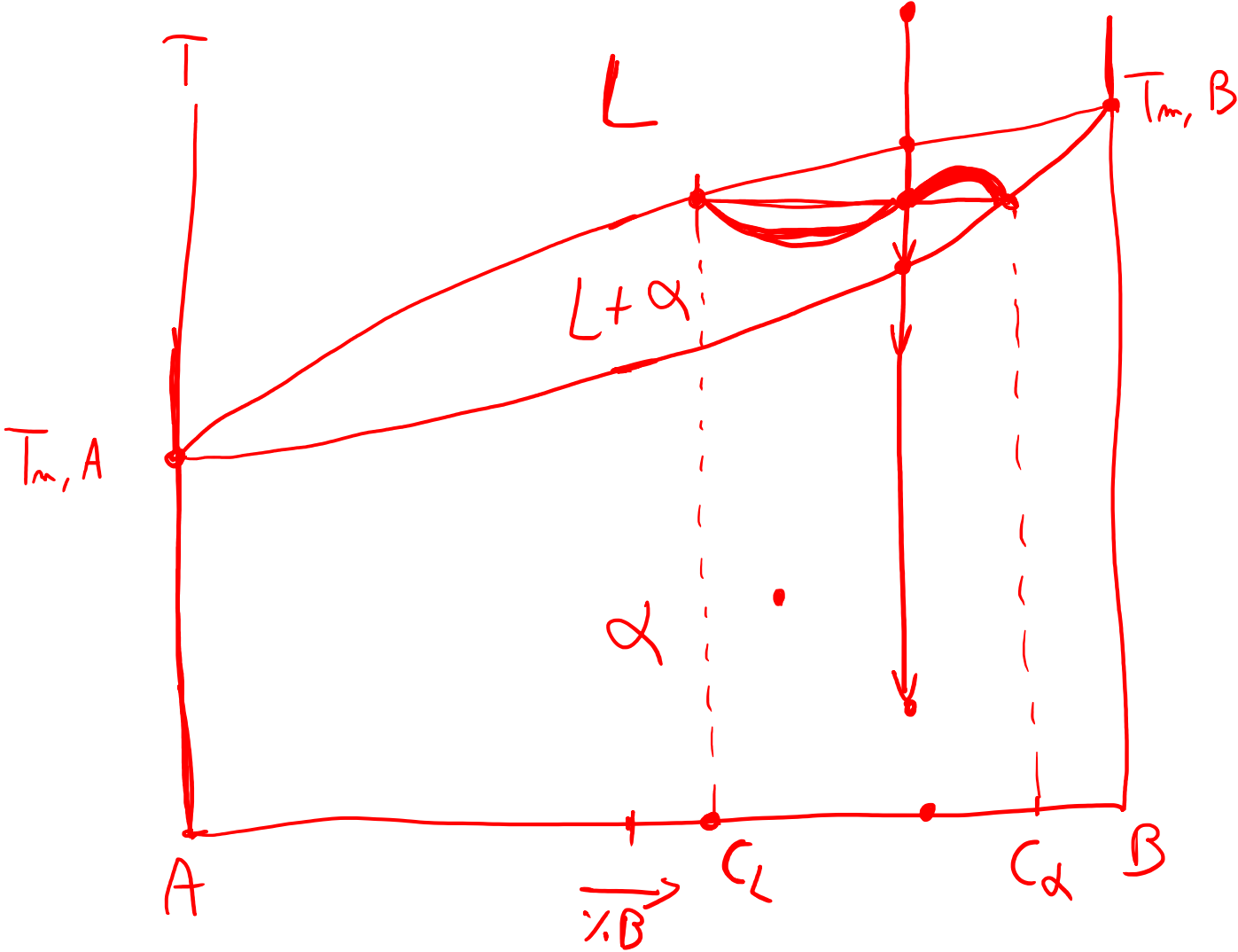
$$\text{Fraction of phase 1} = \frac{(C_2 - C)}{(C_2 - C_1)}$$

$$\text{Fraction of phase 2} = \frac{(C - C_1)}{(C_2 - C_1)}$$

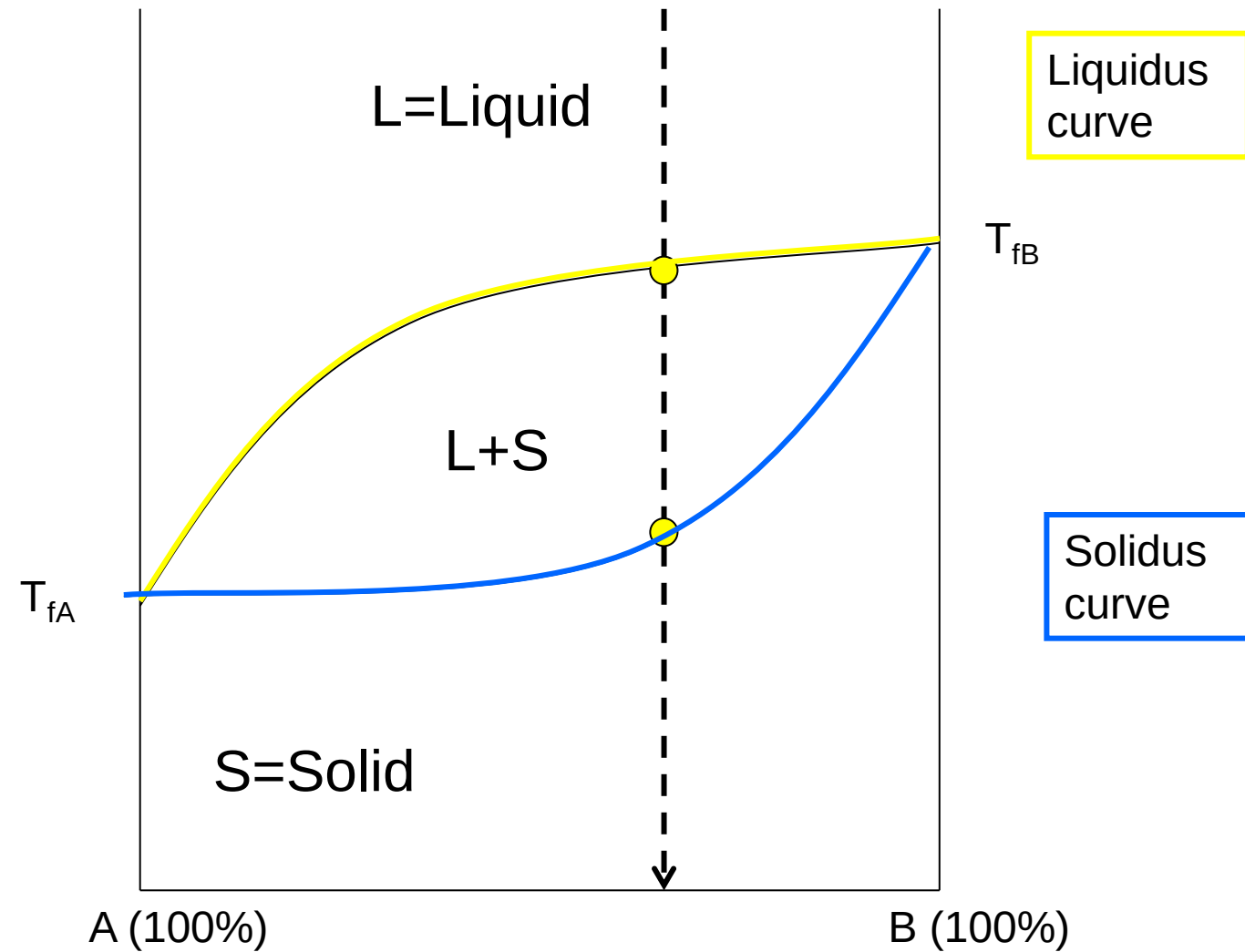


Exercise 1: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **complete solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

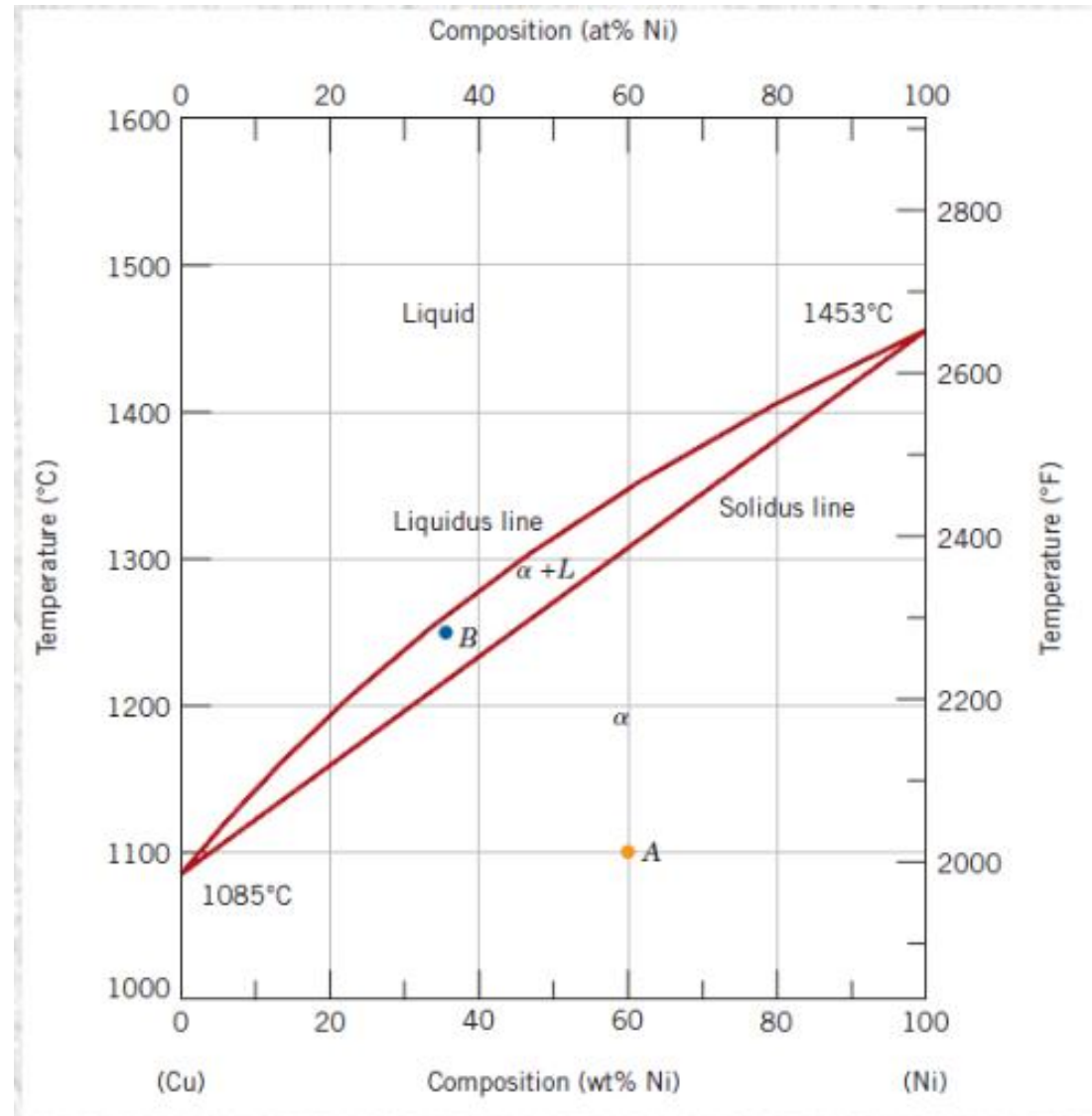
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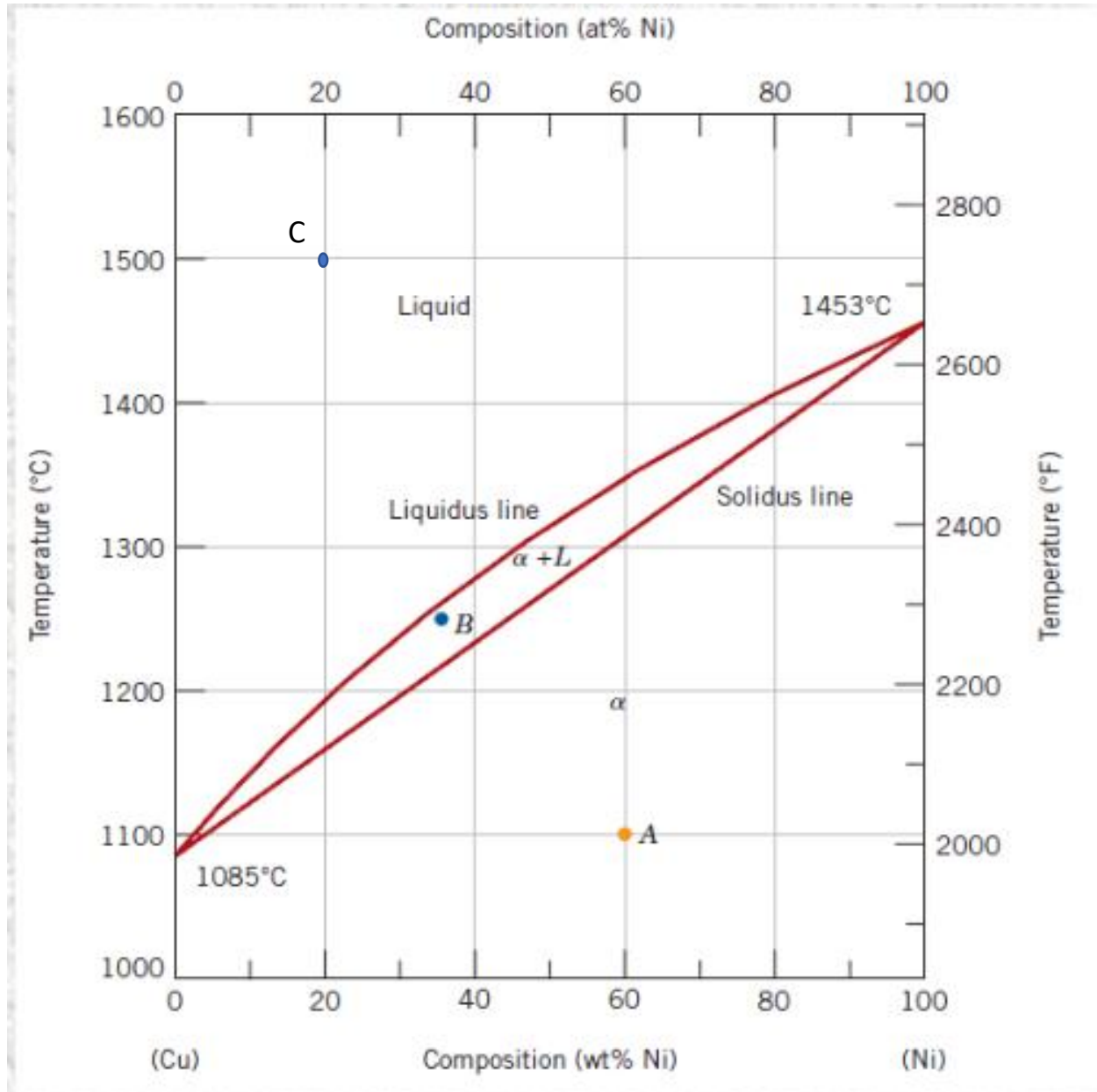
FULL SOLUBILITY IN THE SOLID AND LIQUID STATE



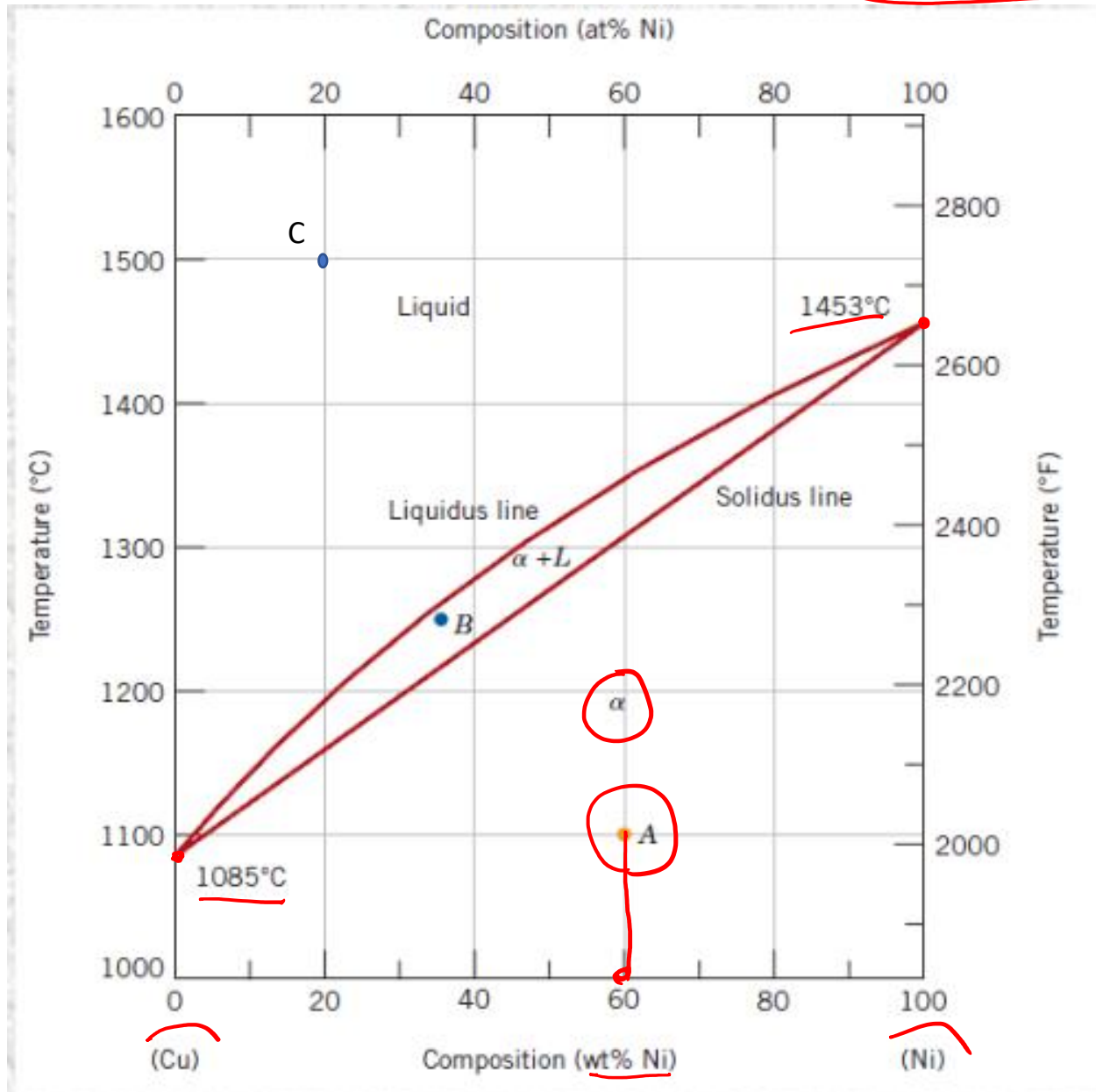
Exercise 1: full solubility in the solid state



Exercise 2: determine the phases present, their composition and relative amounts in the point A, B and C indicated in the phase diagram. Schematize the microstructure in each point.



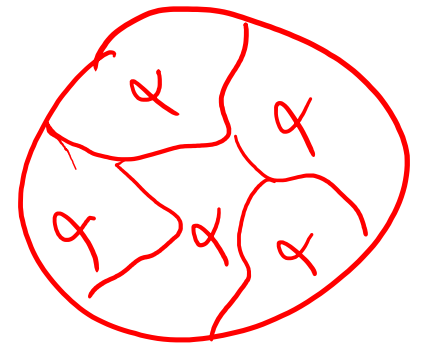
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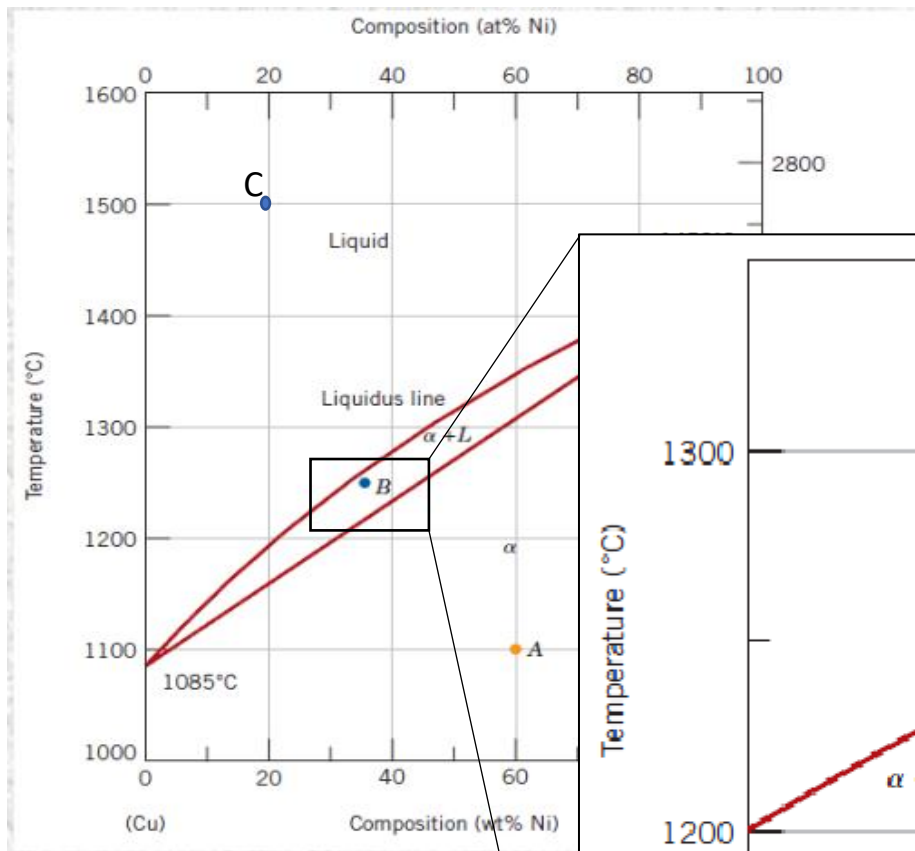


Phase(s) $\rightarrow \alpha$

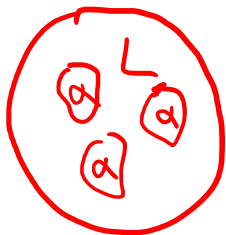
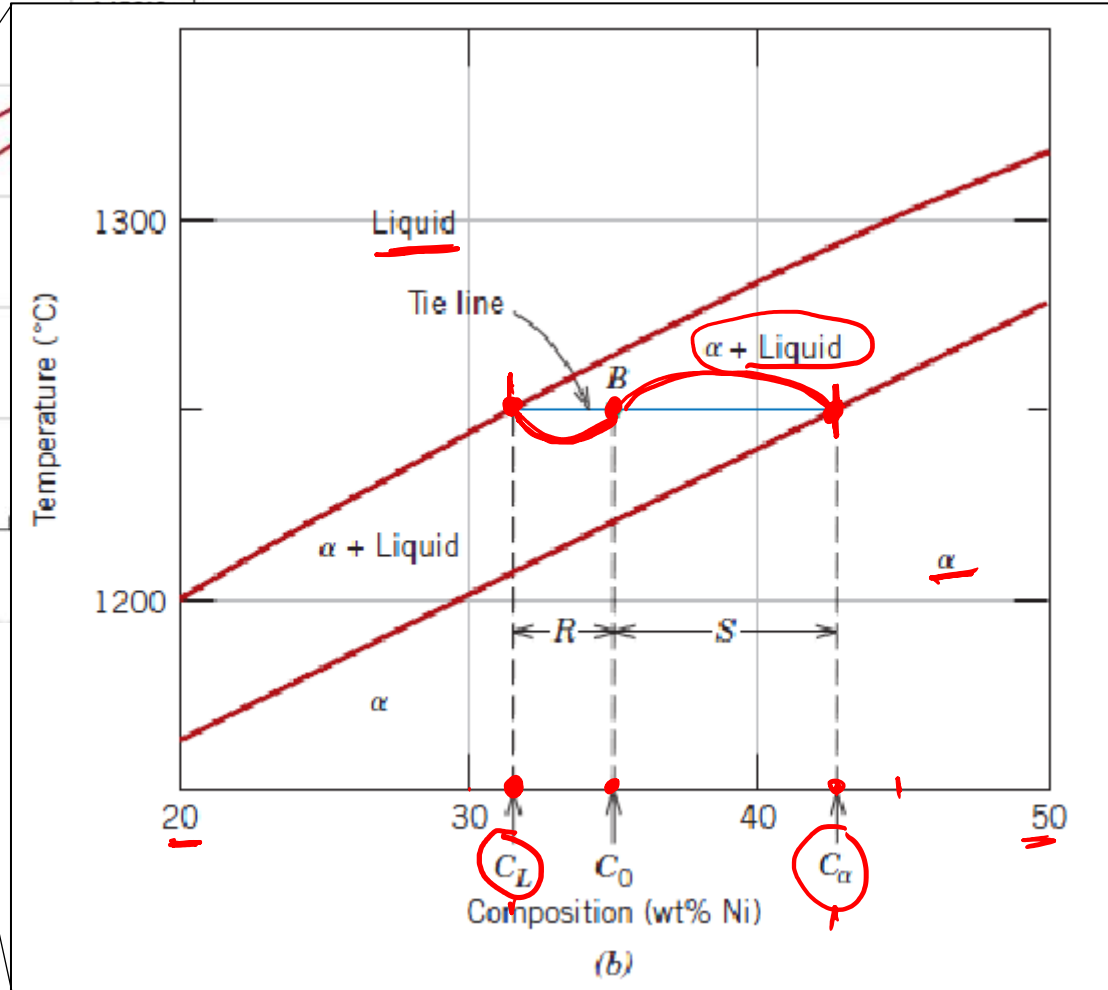
$$C_{\alpha} = \begin{cases} 60\% \text{ Ni} \\ 40\% \text{ Cu} \end{cases}$$

% A $\rightarrow 100\% \alpha$

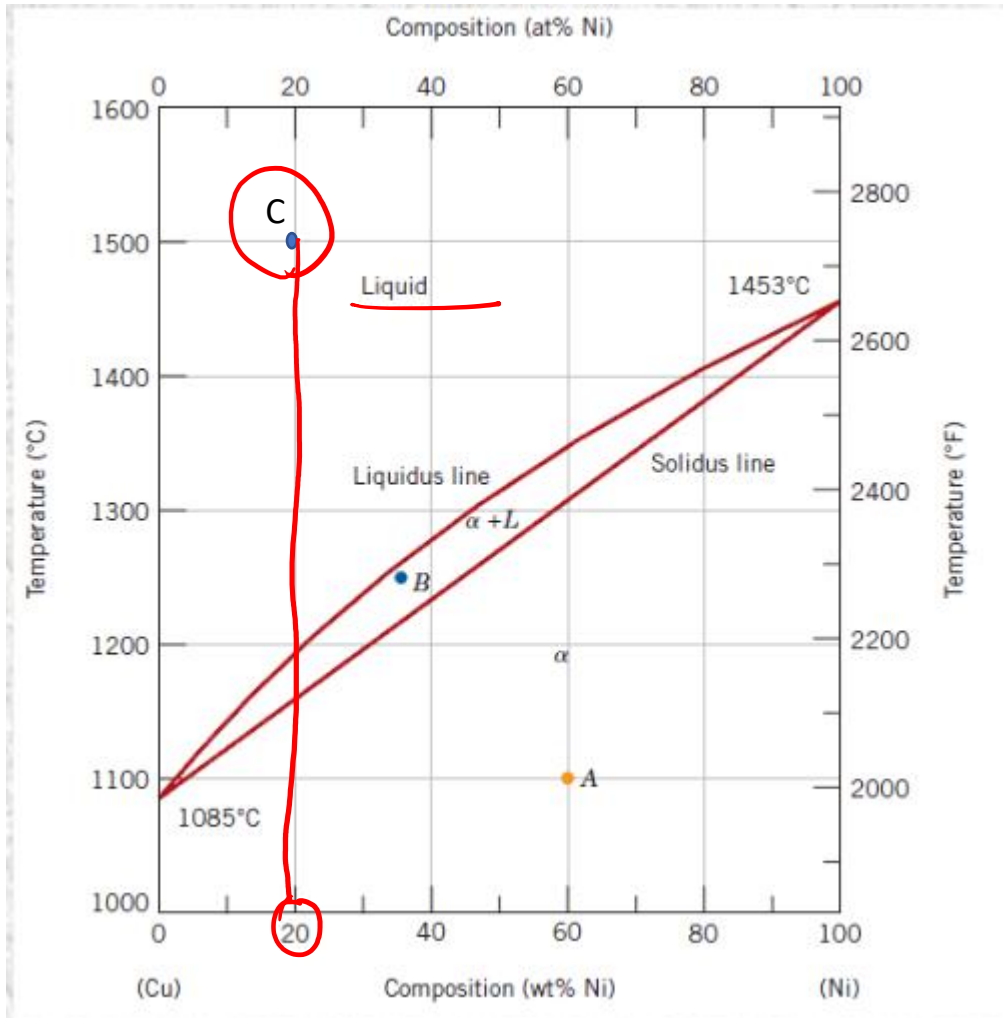




$P_R \rightarrow \underline{\alpha + L}$



	$\downarrow$	$\downarrow$
(B)	$\hookleftarrow$	$\%$
$\rightarrow \alpha$	$\frac{43\% \text{ Ni}}{57\% \text{ Cu}}$	$\frac{35 - 32}{43 - 32} \cdot 100 = \textcircled{27,3\%}$
$\rightarrow L$	$\frac{32\% \text{ Ni}}{68\% \text{ Cu}}$	$\frac{43 - 35}{43 - 32} \cdot 100 =$
		$= \textcircled{72,7\%}$

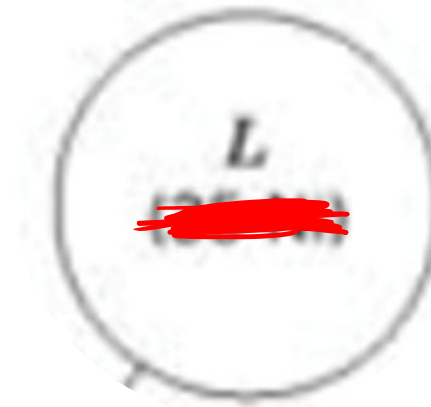


Point C:

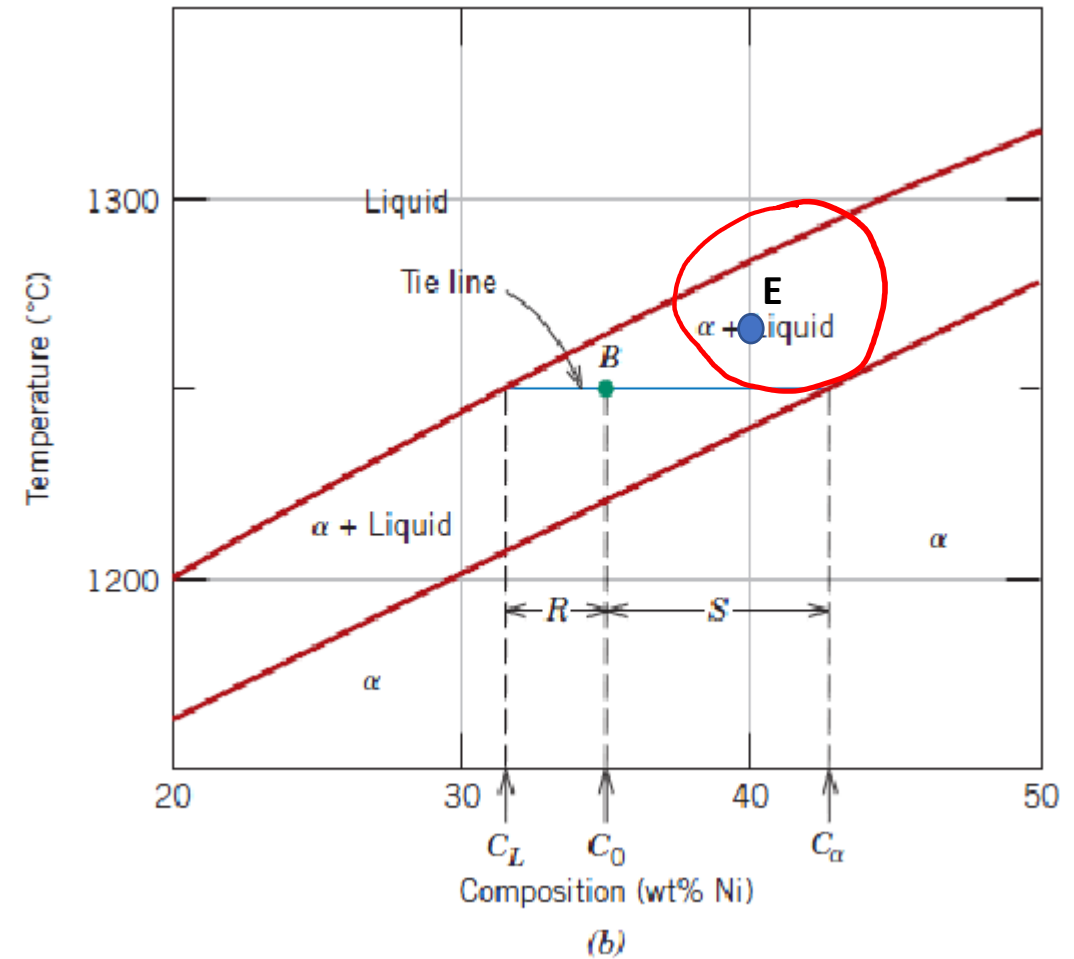
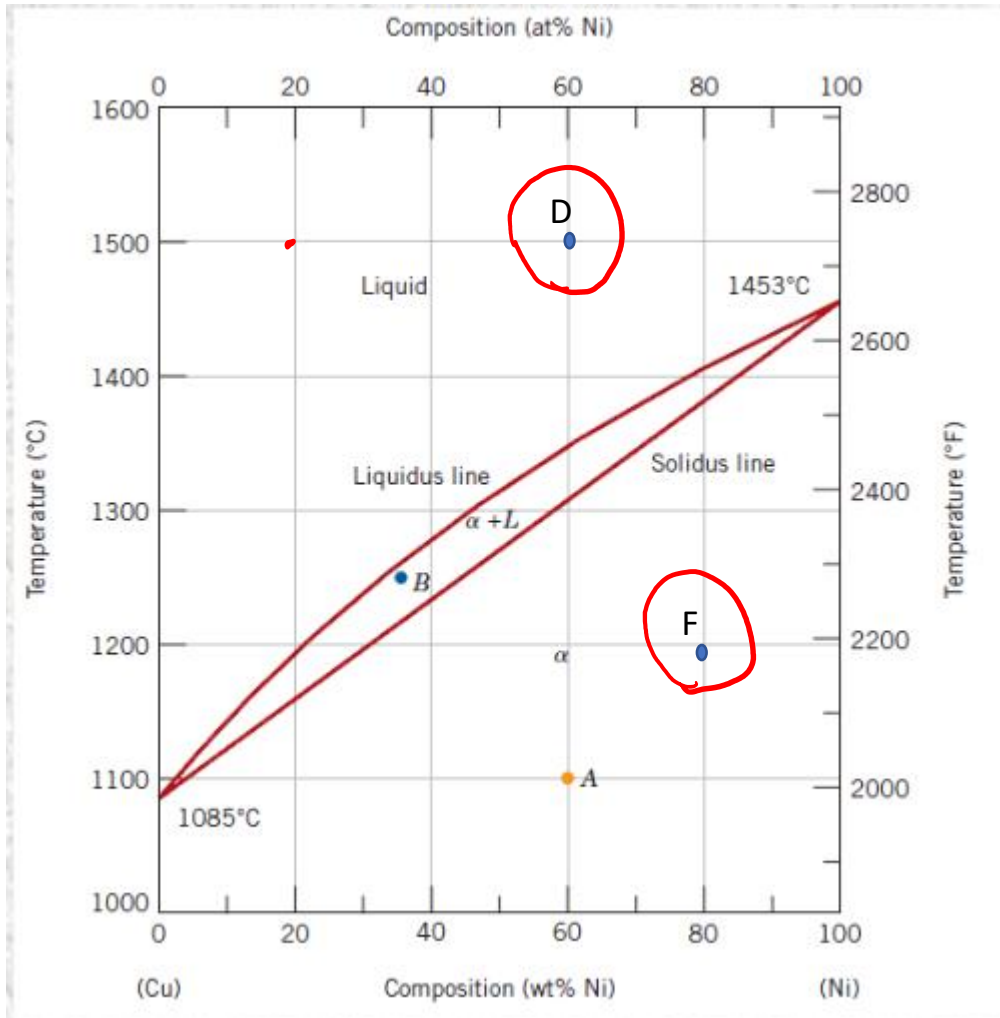
Phases: 1 phase liquid

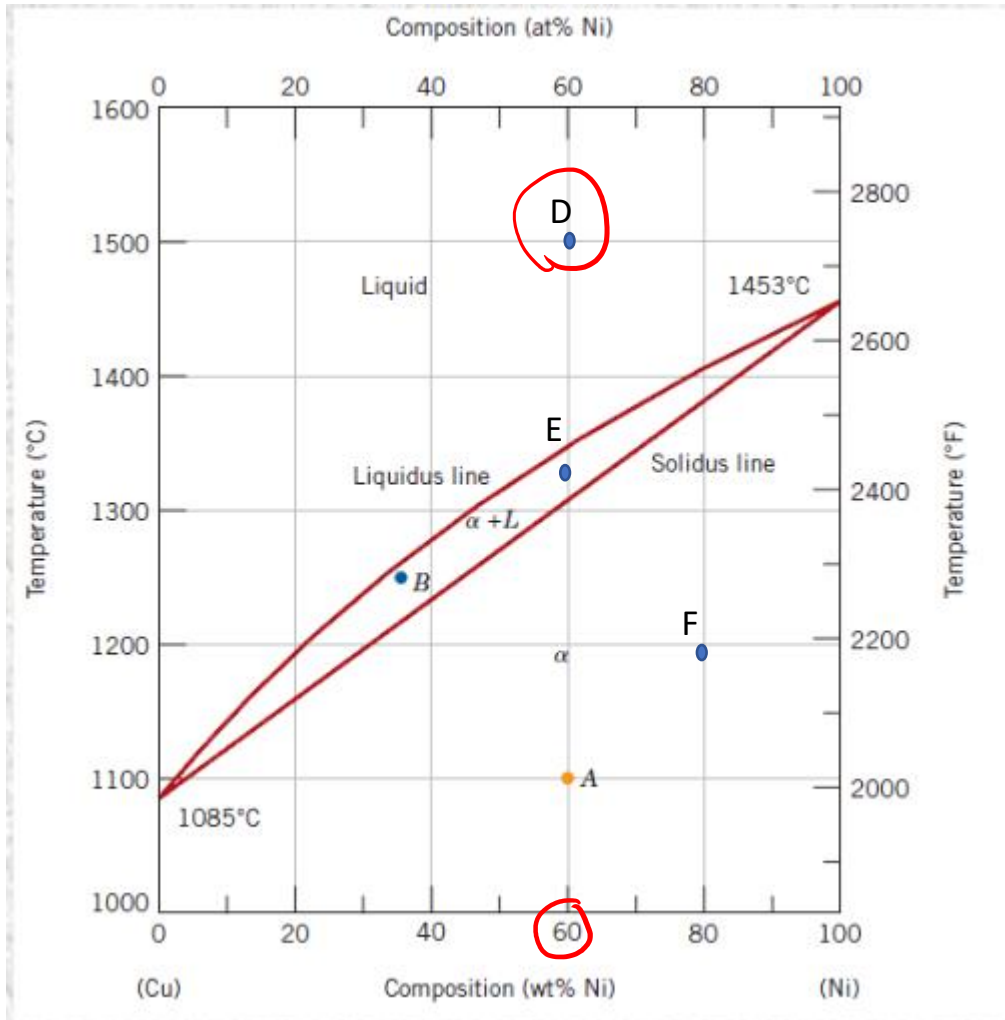
Composition: 20%Ni, 80% Cu

Relative amount: 100% liquid phase



Exercise 3: determine the phase present, their composition and relative amounts in the point D, E and F indicated in the phase diagram. Schematize the microstructure in each point.

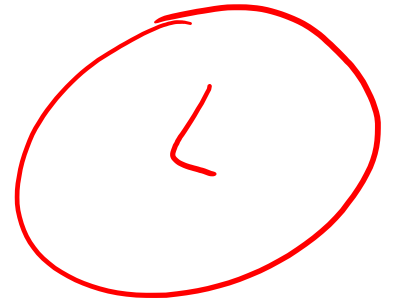


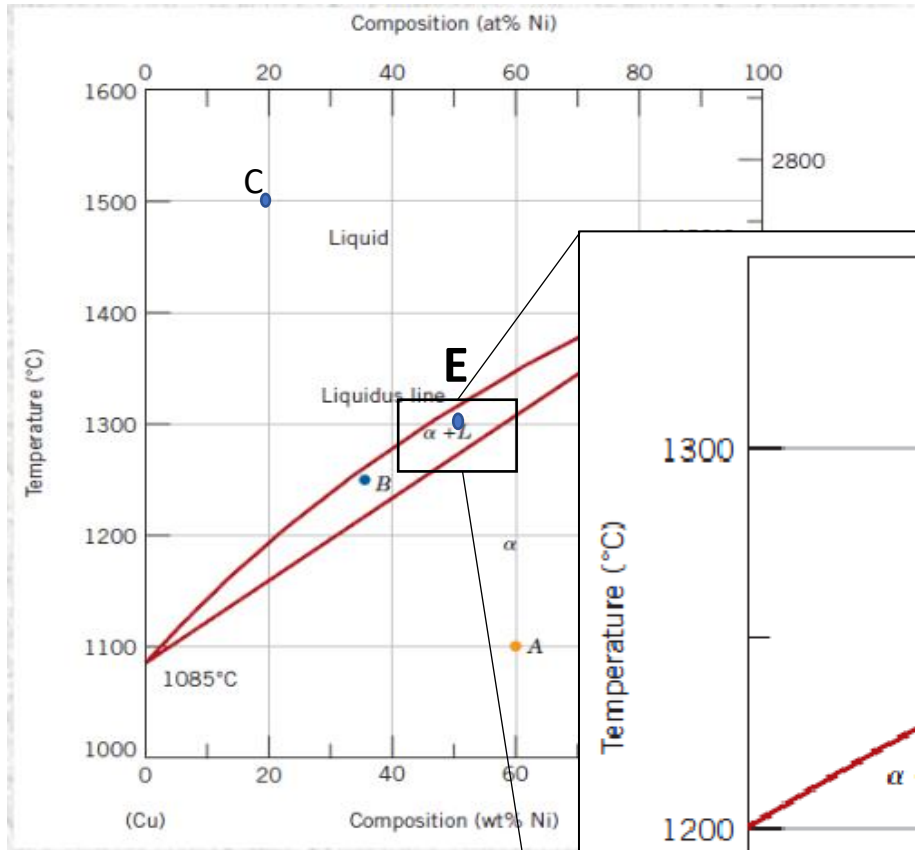


$P_L \rightarrow L$

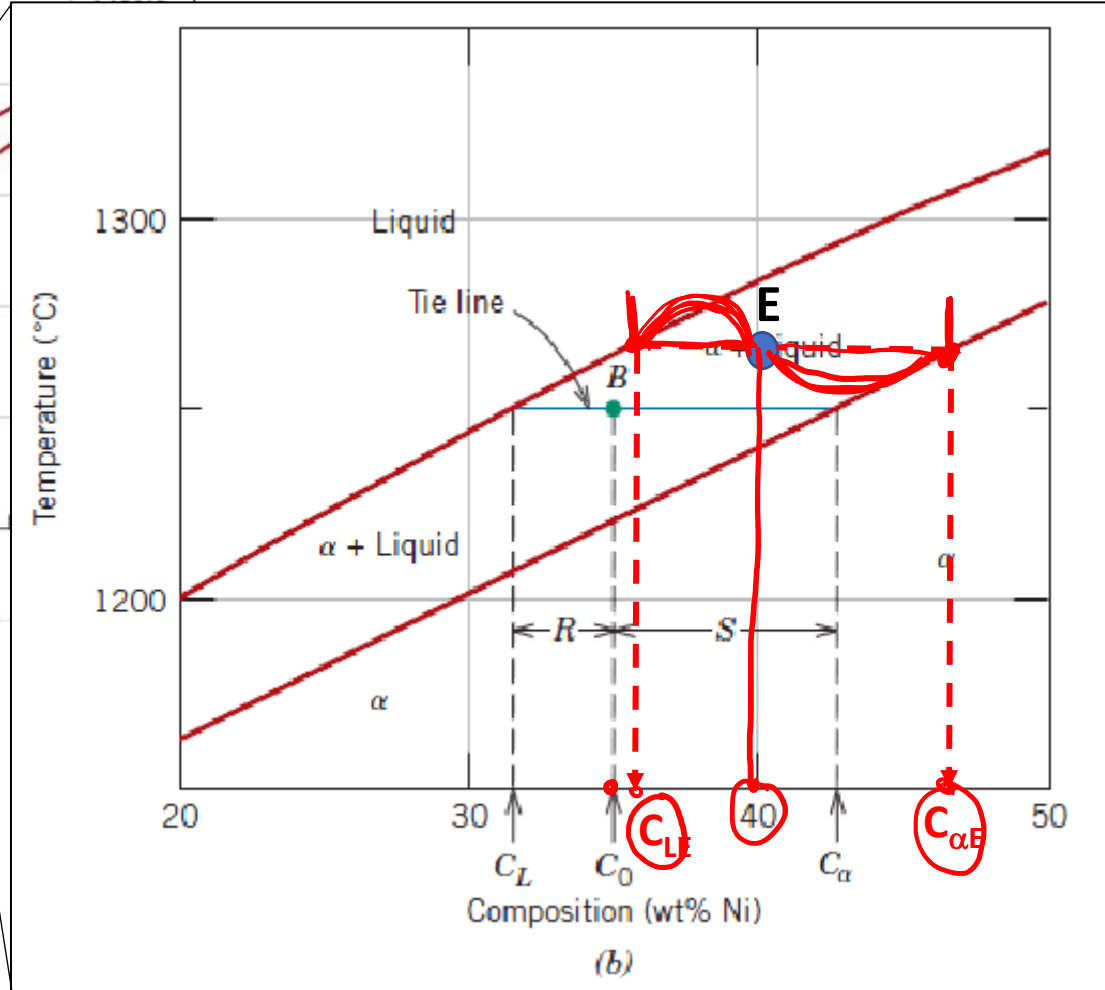
$$C_L = \begin{cases} 60\% \text{ Ni} \\ 40\% \text{ Cu} \end{cases}$$

$\therefore D \rightarrow 100\% L$

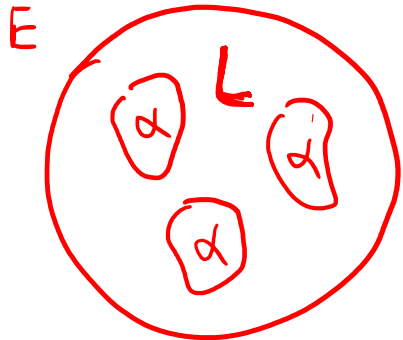


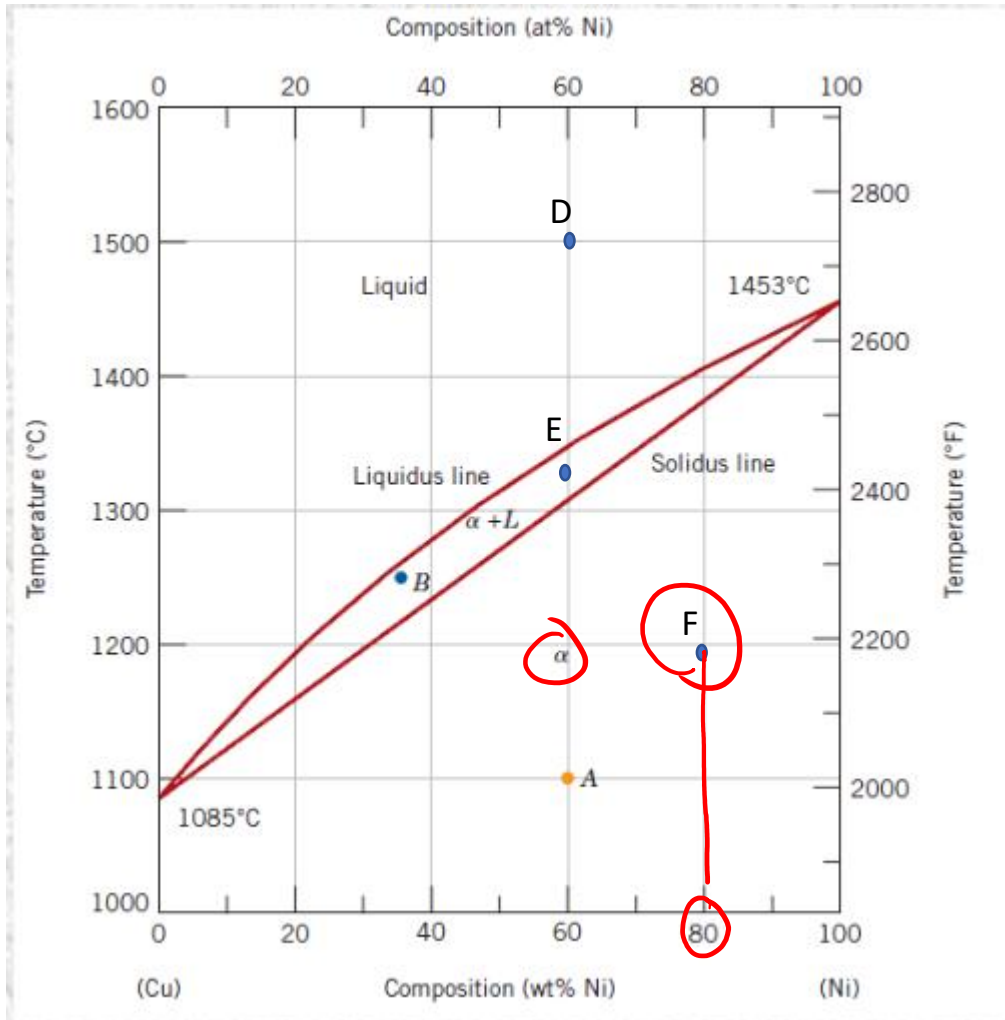


Ph $\rightarrow$ $\alpha + L$



(E)	C	%
α	47% Ni 53% Cu	$\frac{40-36}{47-36} \cdot 100 = 36\%$
L	36% Ni 64% Cu	$\frac{47-40}{47-36} \cdot 100 = 64\%$

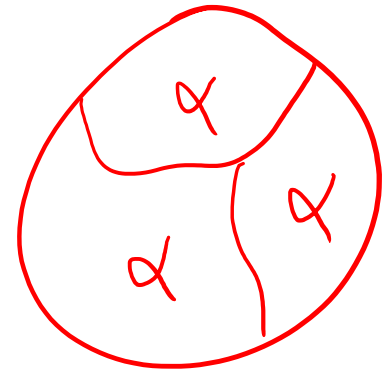




$P_h \rightarrow \alpha$

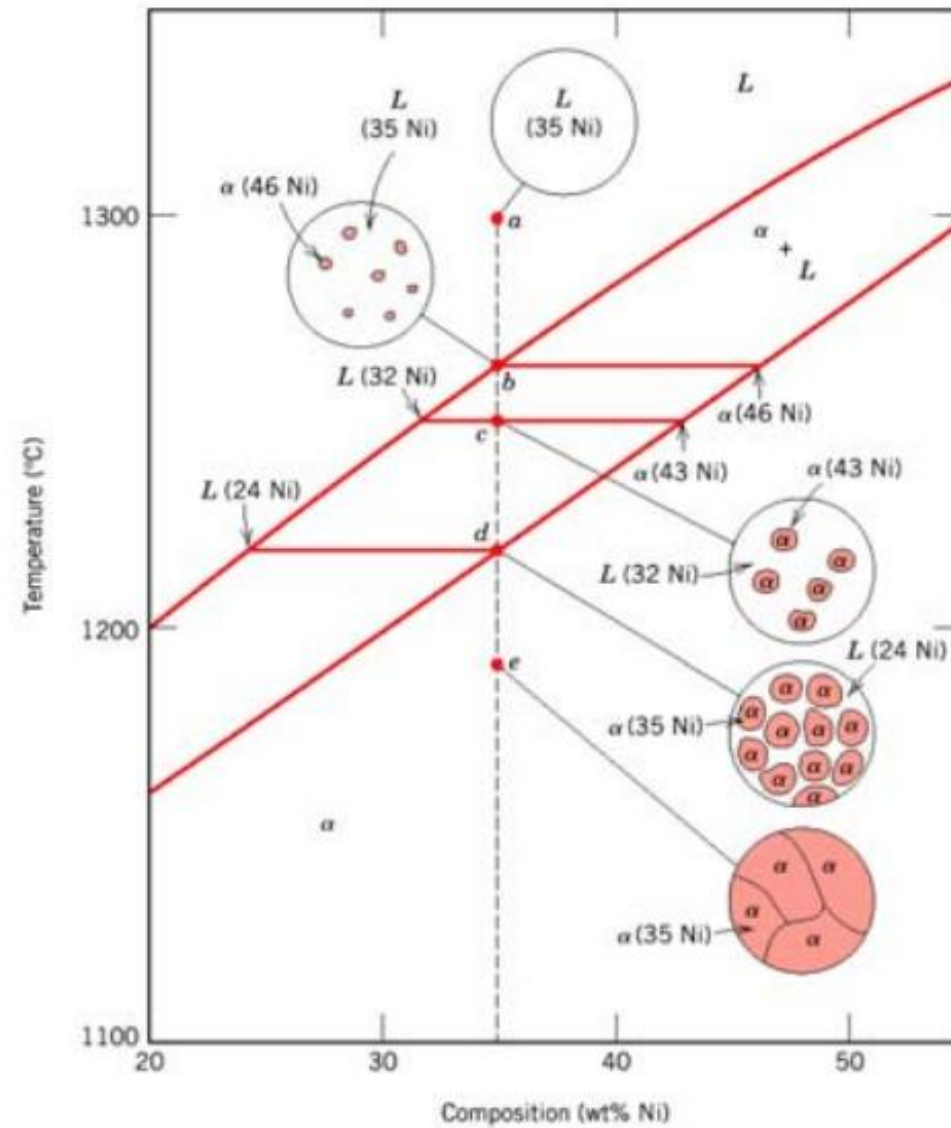
$C_\alpha \begin{cases} 80\% \text{ Ni} \\ 20\% \text{ Cu} \end{cases}$

$\%F \rightarrow 100\% \alpha$



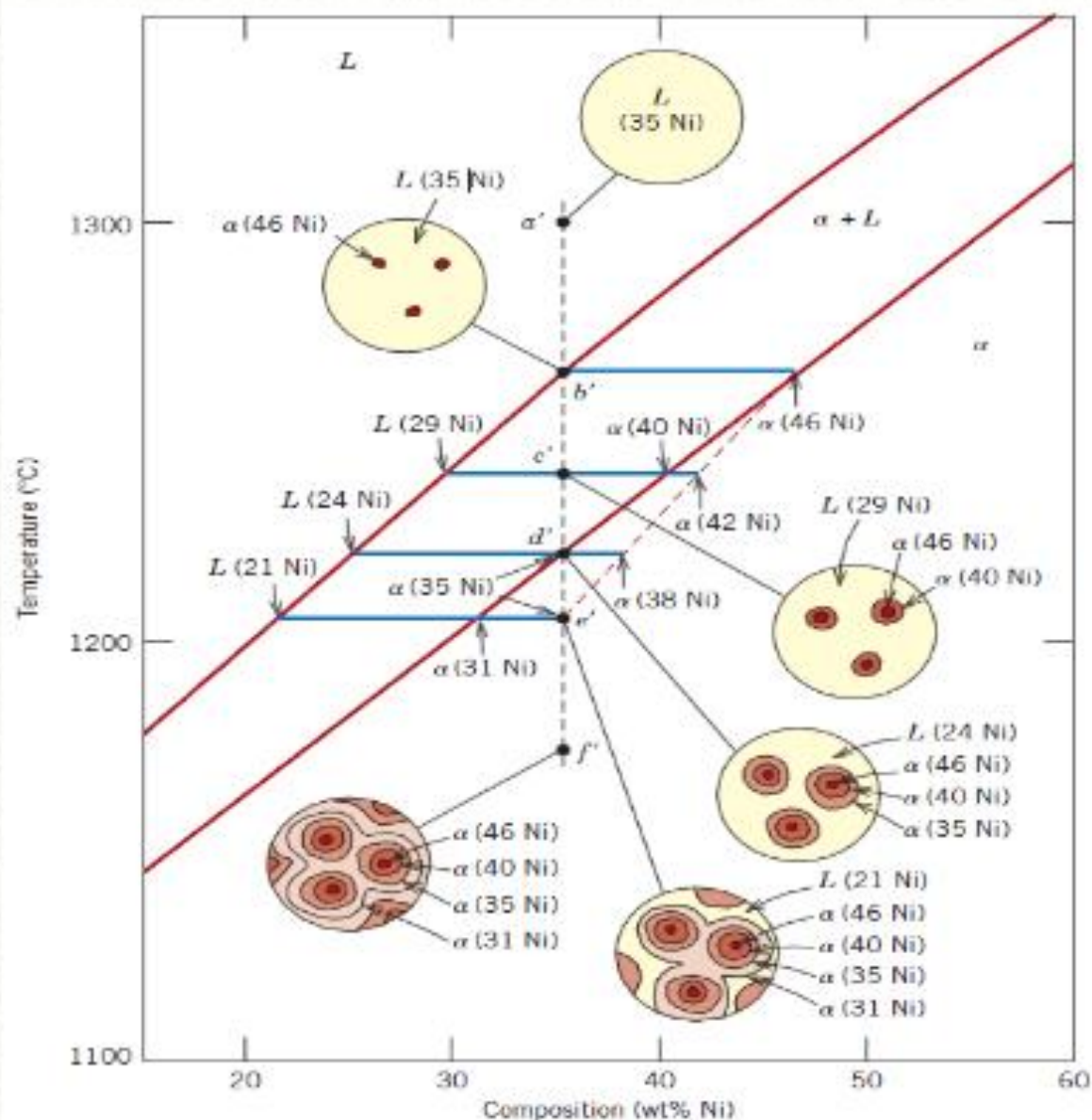
Development of microstructure in isomorphous alloys

Equilibrium (very slow) cooling

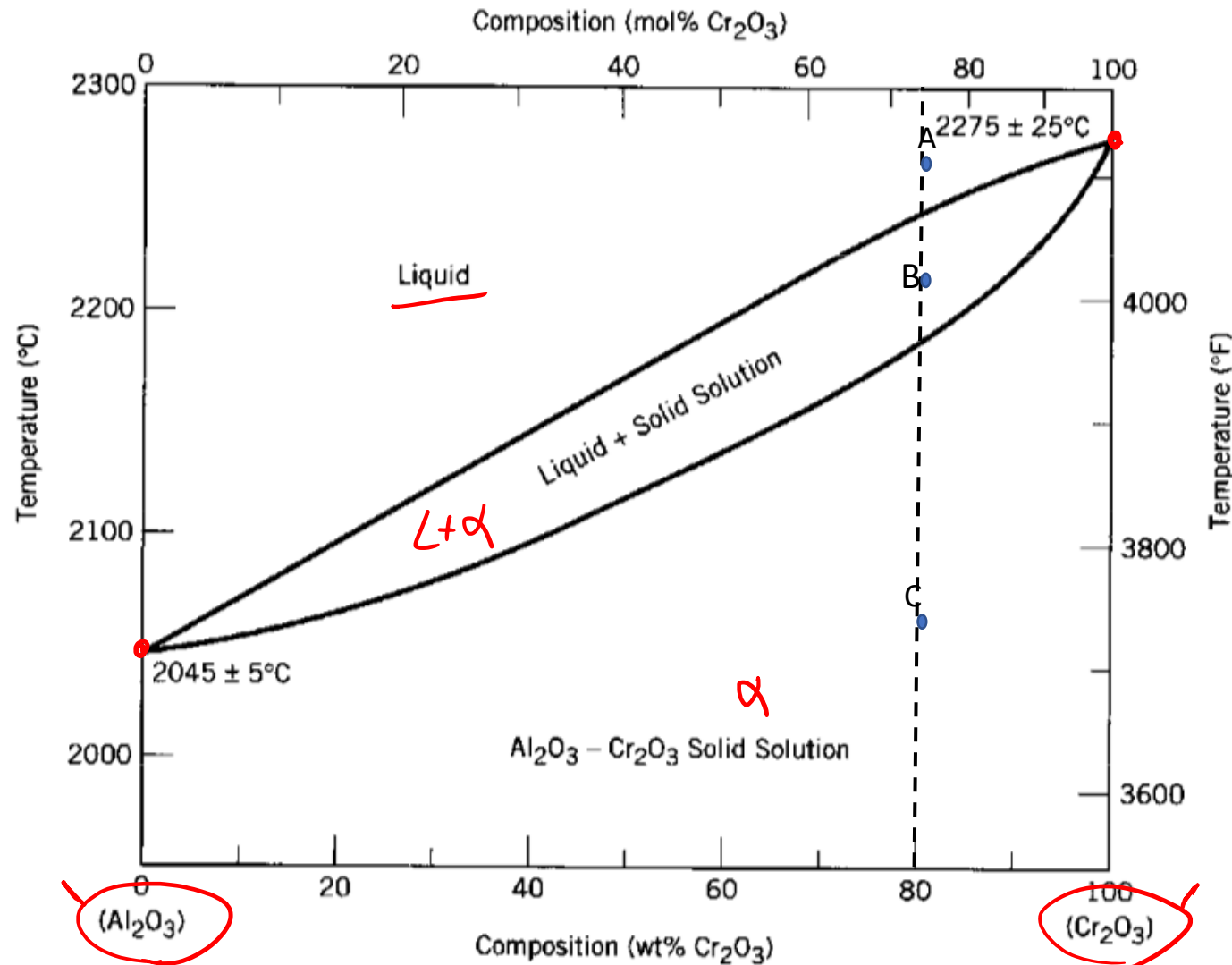


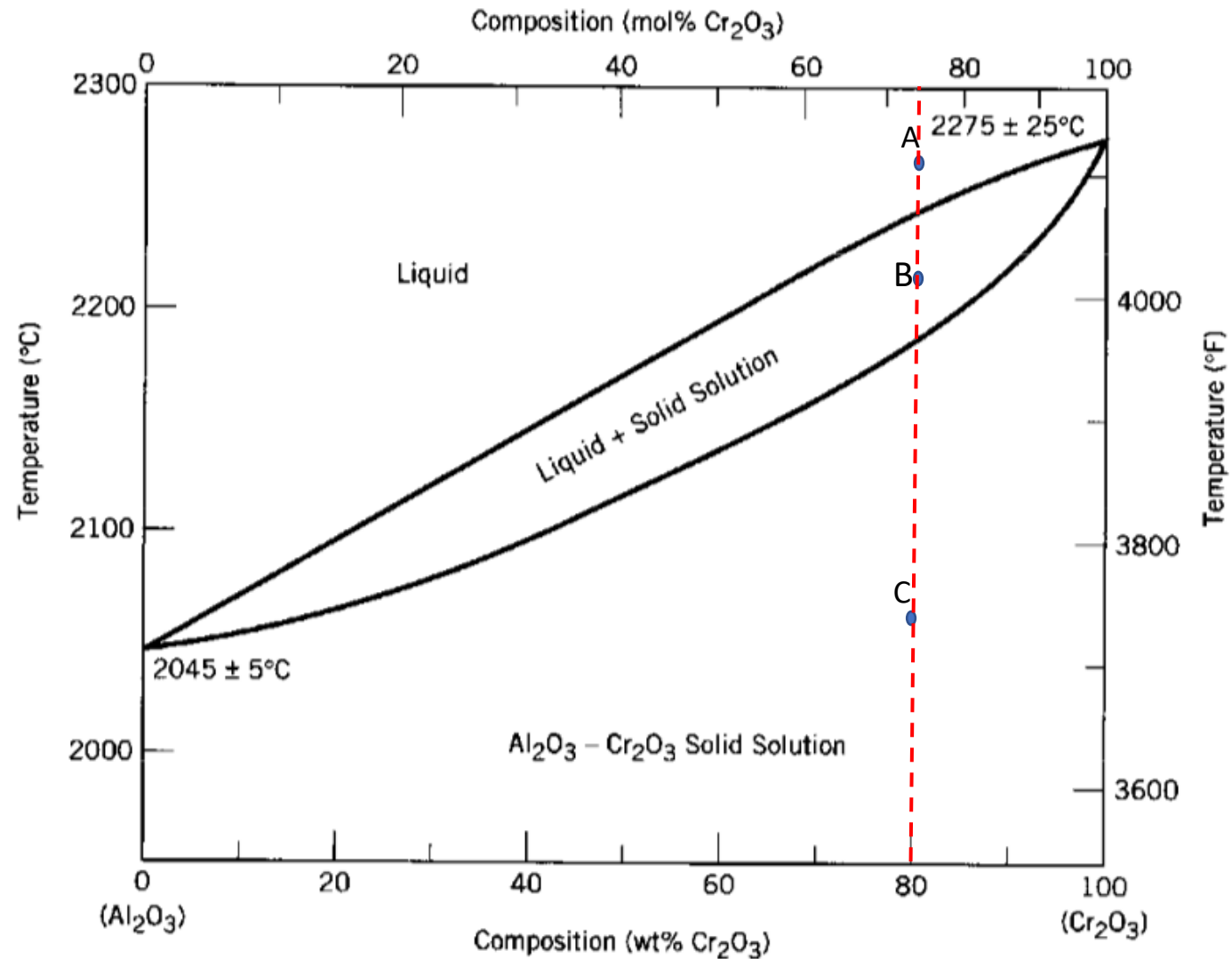
Non-Equilibrium Cooling

For Large cooling rate there is not enough time for diffusion!



Exercise 4: determine the phase present, their composition and relative amounts in the point A, B and C indicated in the phase diagram.



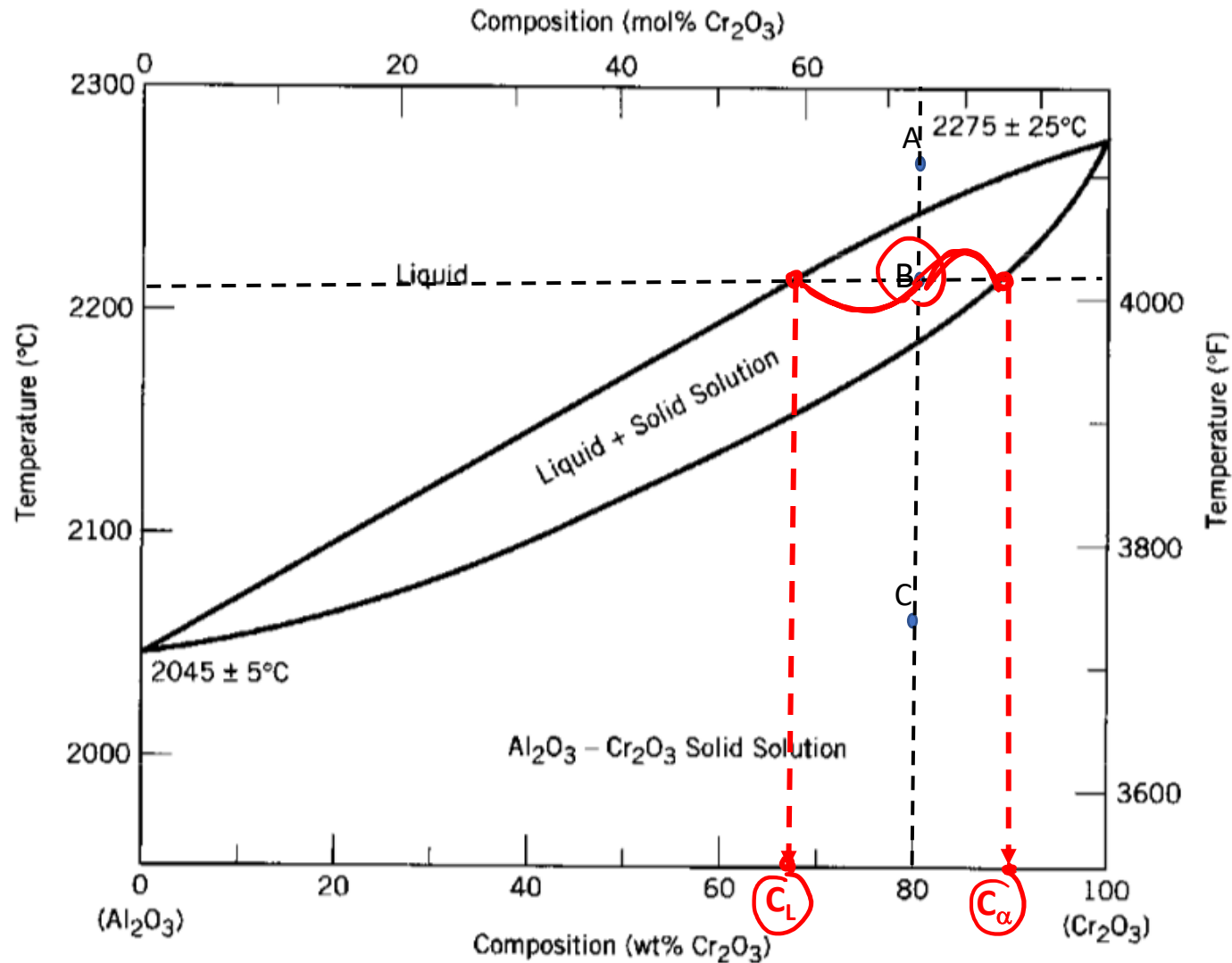


Point A:

Phases: 1 phase liquid

Composition: 80% Cr_2O_3 , 20% Al_2O_3

Relative amount: 100% liquid phase



(B)	C	%
L	68% Cr_2O_3 32% Al_2O_3	45.5%
α	90% Cr_2O_3 10% Al_2O_3	54.5%

Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition:

2 phases region $\rightarrow$ horizontal rule

Composition of liquid (L):

$C_L = 68\% \text{Cr}_2\text{O}_3, 32\% \text{Al}_2\text{O}_3$

Composition of solid (α):

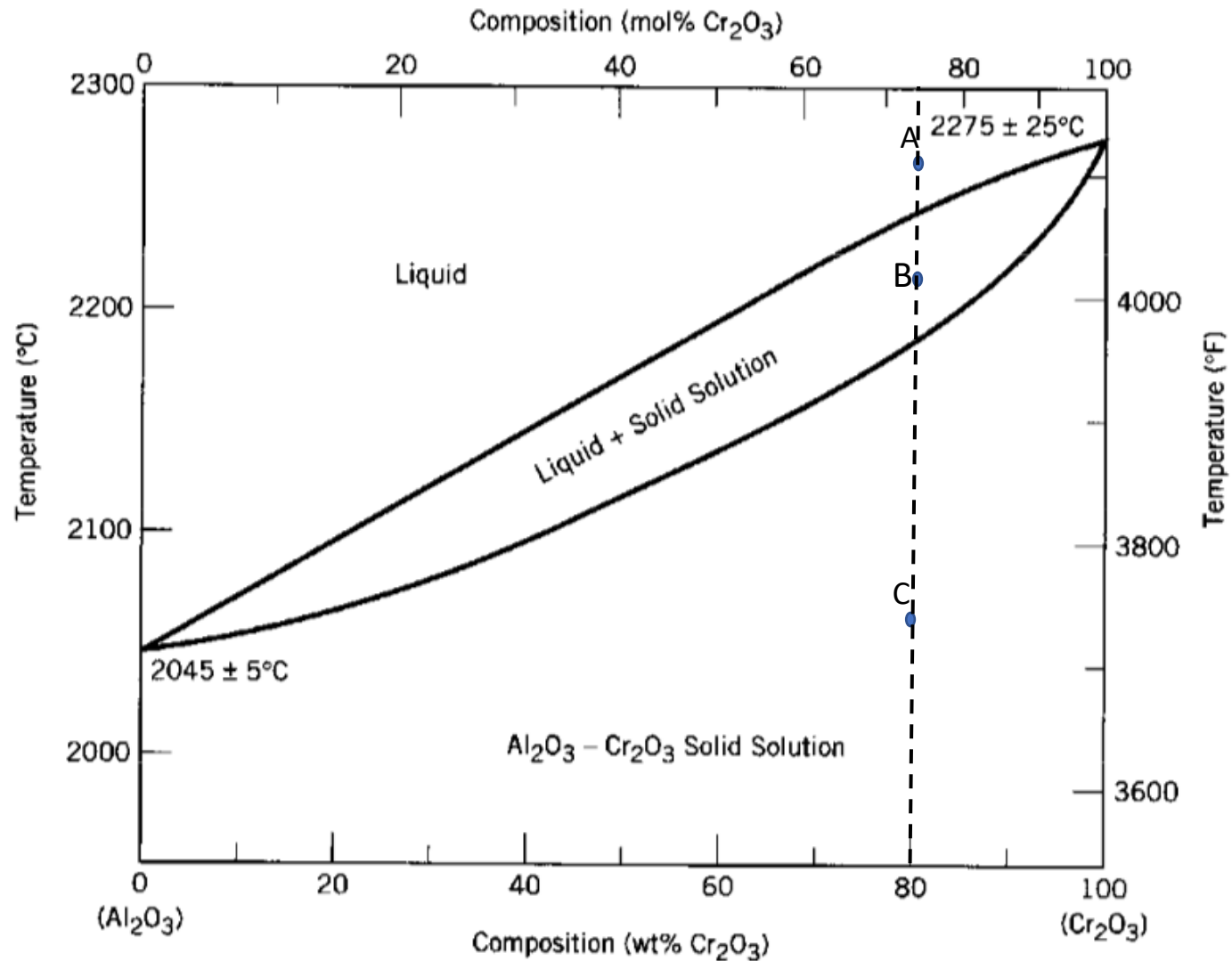
$C_{\alpha} = 90\% \text{Cr}_2\text{O}_3, 10\% \text{Al}_2\text{O}_3$

Relative amount:

2 phases region $\rightarrow$ lever rule

$$\% \alpha = (80 - 68) / (90 - 68) * 100 = 54.5\%$$

$$\% L = (90 - 80) / (90 - 68) * 100 = 45.5\%$$



Point C:

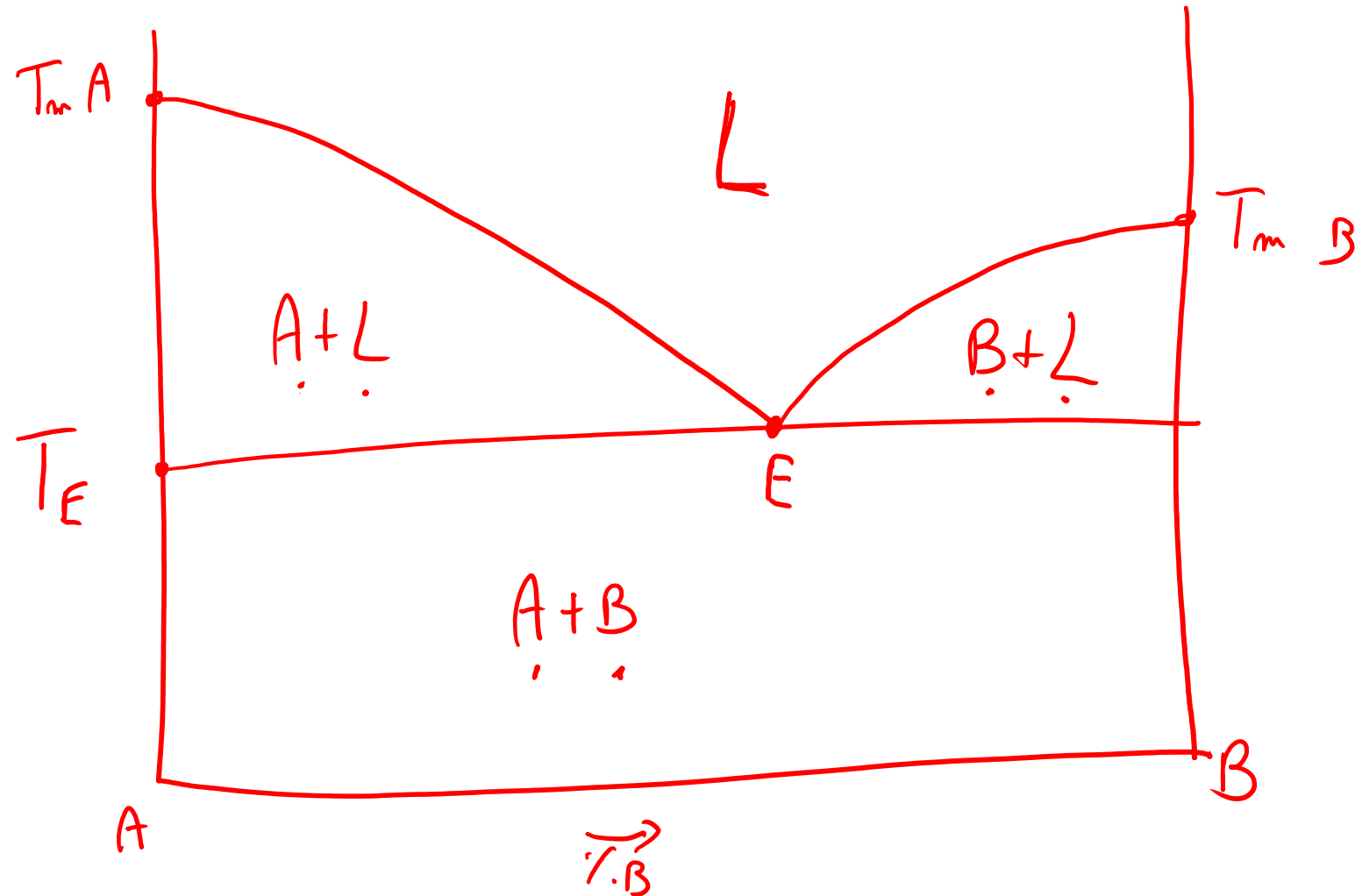
Phases: 1 phase solid

Composition: 80% Cr_2O_3 , 20% Al_2O_3

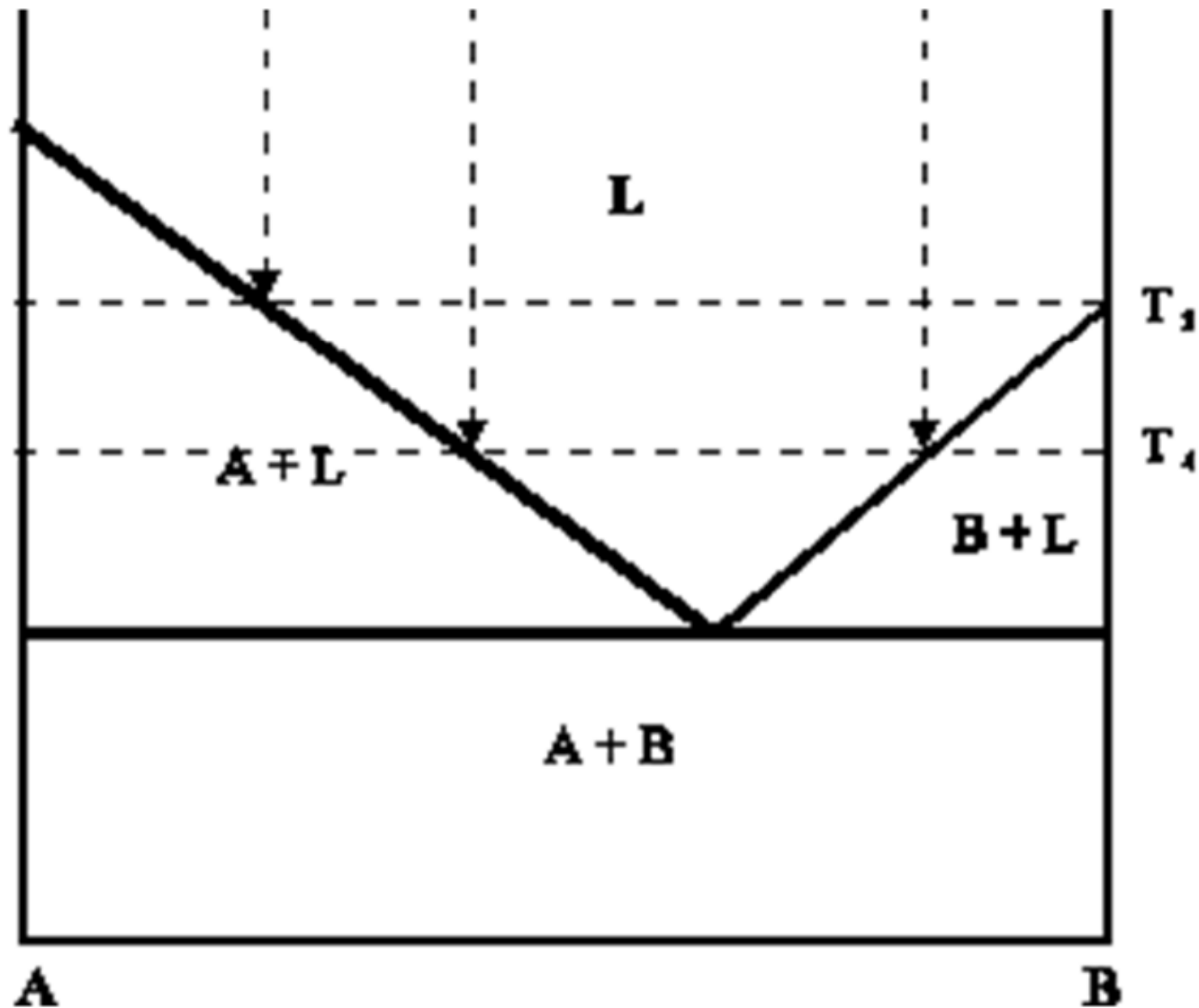
Relative amount: 100% solid phase

Exercise 5: Draw a phase diagram (for generic components A and B), with **complete solubility in the liquid state** and **no solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

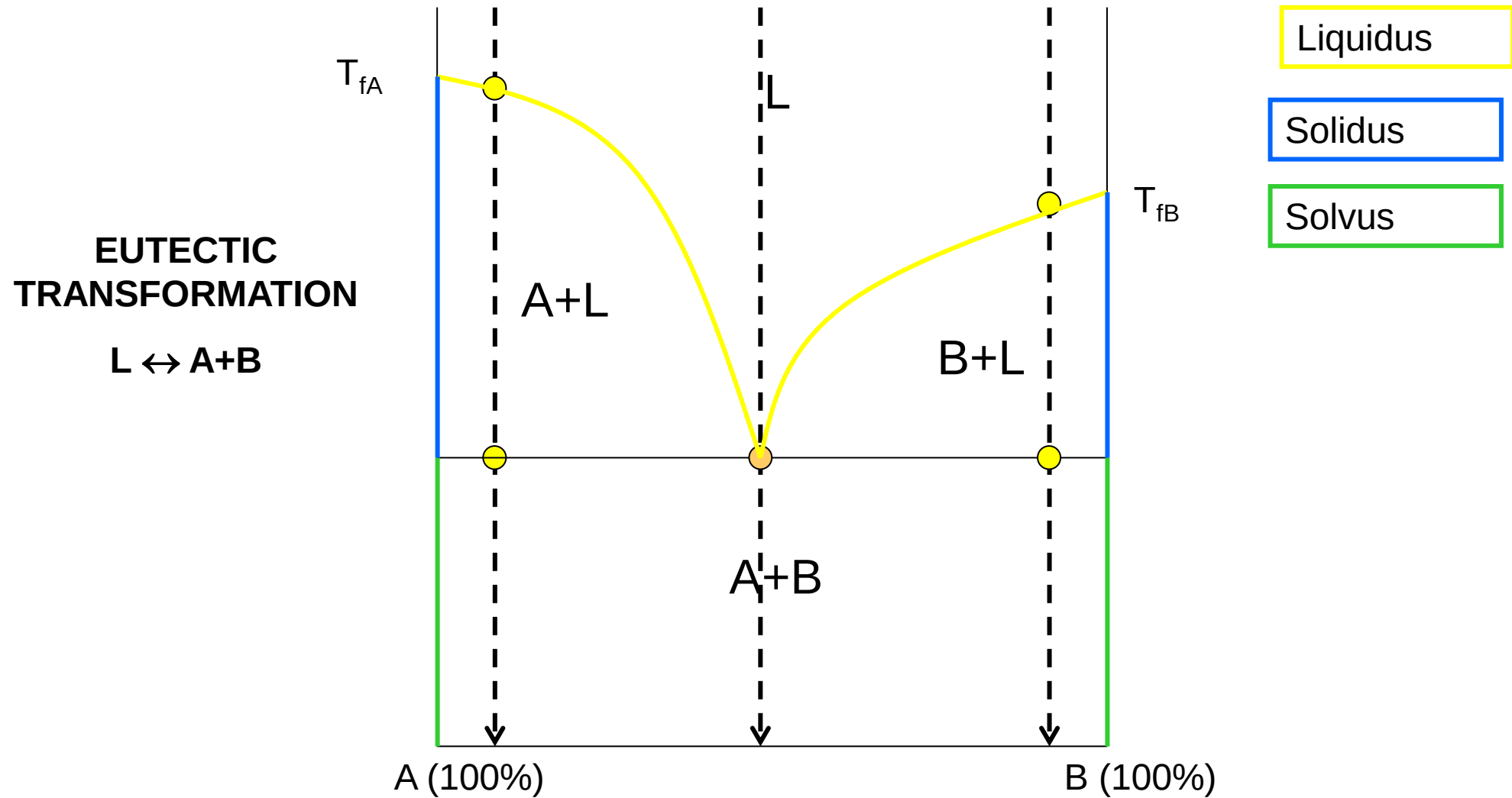
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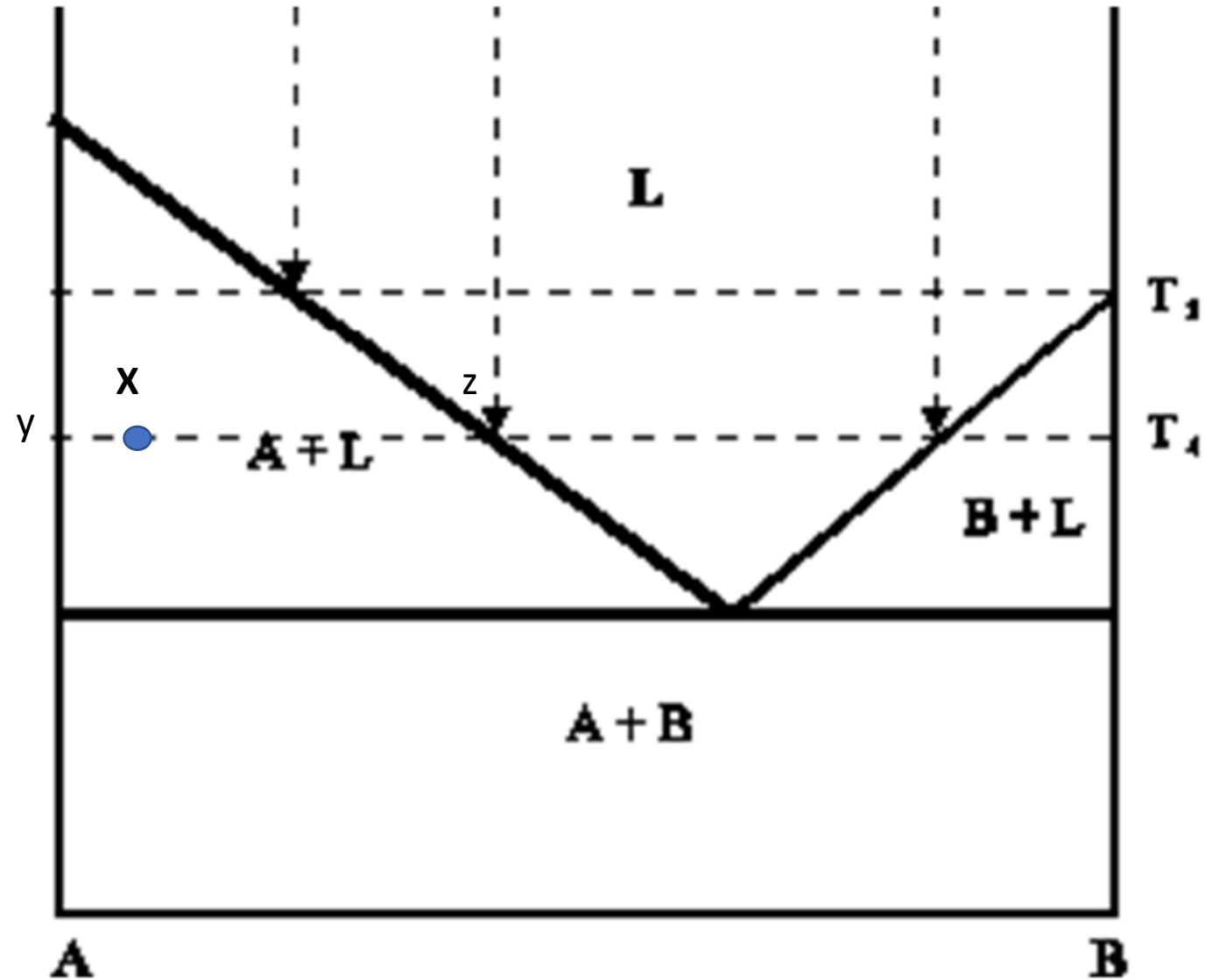
Complete solubility in the liquid state, **no solubility in the solid state**



COMPLETE SOLUBILITY IN THE LIQUID STATE AND NO SOLUBILITY IN THE SOLID STATE, EUTECTIC



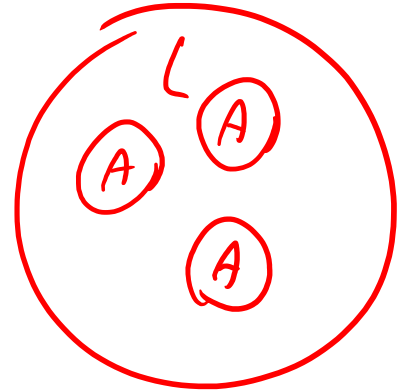
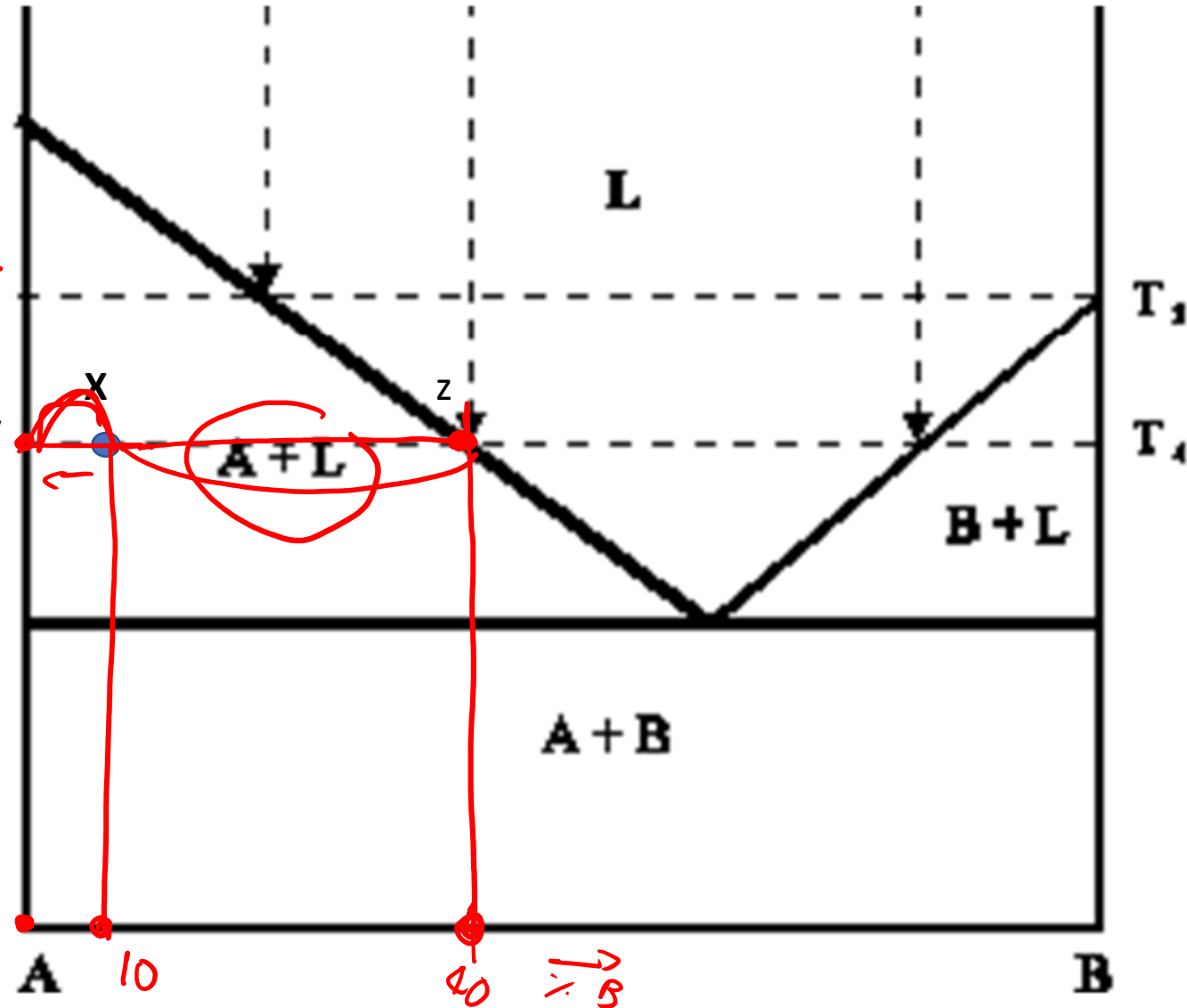
Exercise 6: determine the phases present, their compositions and relative amounts in the point X.



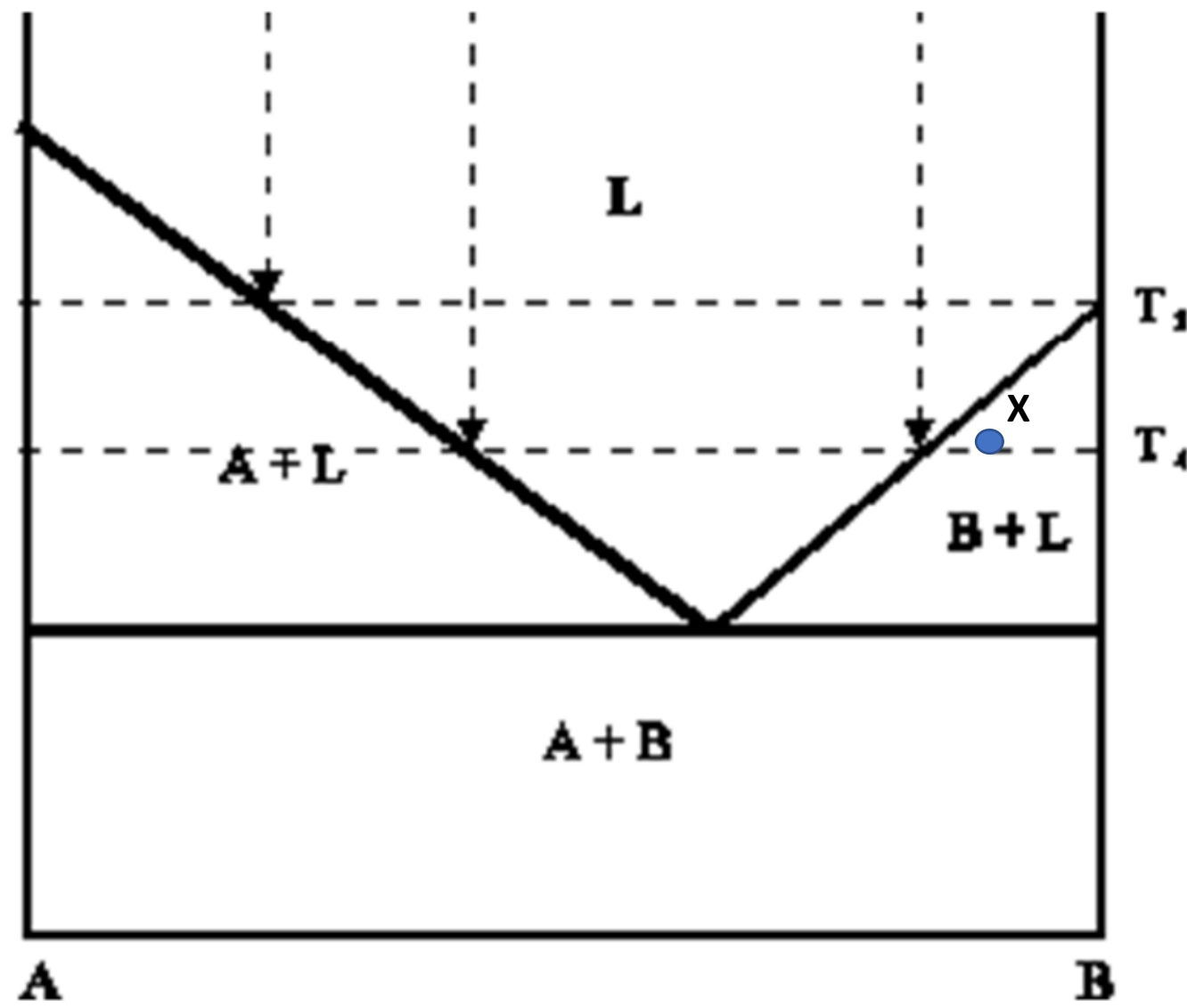
Exercise 6: determine the phases present, their compositions and relative amounts in the point X.

Ph $\rightarrow$ A+L

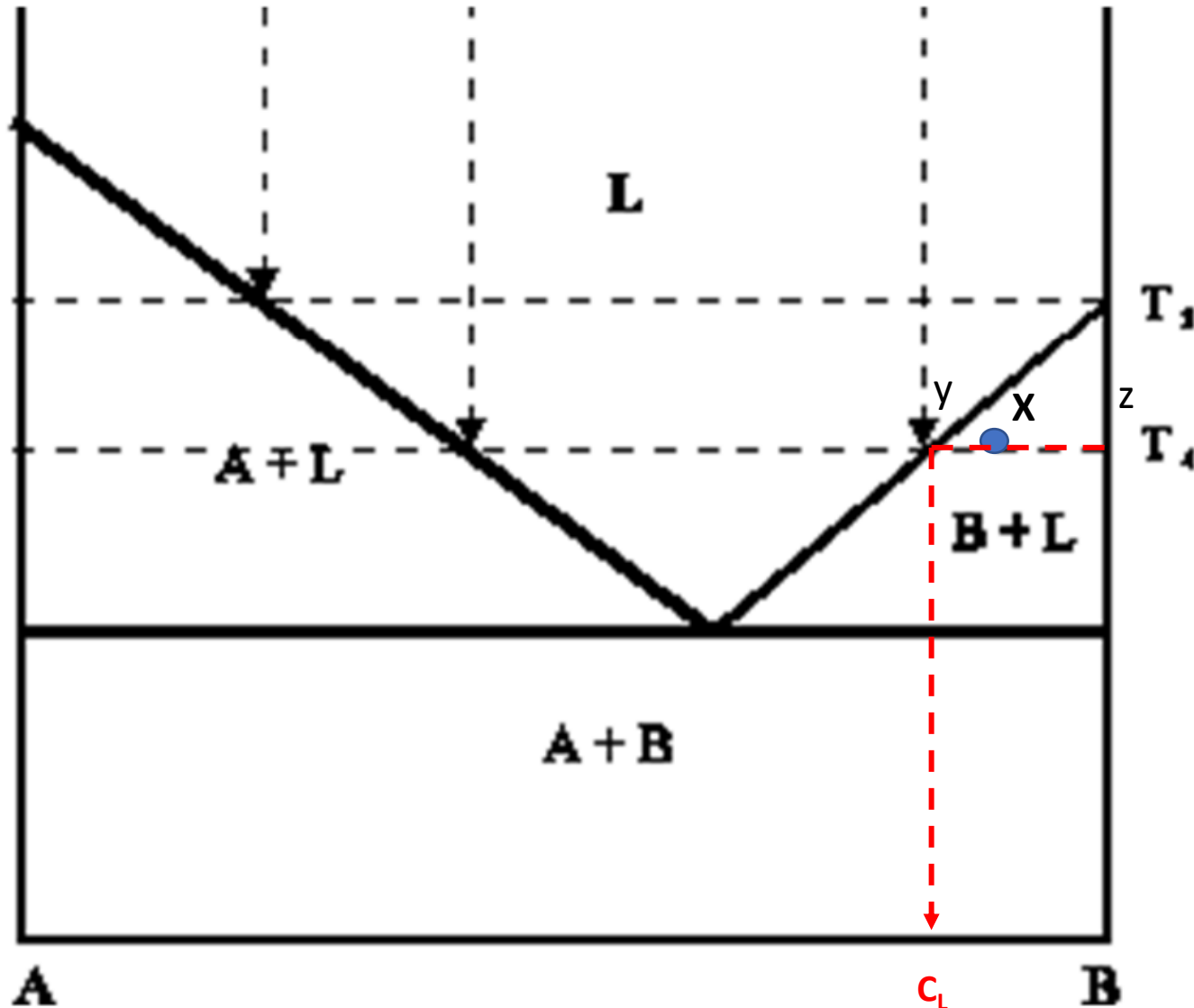
(X)	C	%
A	100% A	$\frac{10-0}{40-0} \cdot 100 = 25\%$
L	40% B 60% A	$\frac{40-10}{40-0} \cdot 100 = 75\%$



Exercise 7: determine the phases present, their compositions and relative amounts in the point X.



Exercise 7: determine the phases present, their compositions and relative amounts in the point X.



Point X

Phases: solid B+ Liquid L

Compositions:

2 phases region → horizontal rule

C_B : 100% B

C_L : 80%B, 20%A

Relative amounts

$$\%B = (Xy)/(yz) * 100 = (85-80)/(100-80) * 100 = 25\%$$

$$\%L = (Xz)/(yz) * 100 = (100-85)/(100-80) * 100 = 75\%$$

